compact_arithmetic21 frozen proof archive

Frozen proof-report archive: compact_arithmetic21

This is a mechanical TeX translation of every bound Markdown proof report, plus prose reports explicitly named by the current proof map. It accompanies the edited manuscript, exact scripts and partial Lean development. It is an archival source collection, not a claim of end-to-end formal verification.

Reports retain their original dates, scopes and historical status wording. Later reports can supersede earlier bounds. The authoritative endpoint is results/fresh_arithmetic_dimension21_composition_20261016_bound21.json. The current edited proof map is `docs/compact_arithmetic21_proof_map.md`.

The manifest records all report hashes and every recursively bound file. This export verifies source identity; it does not reprove the geometry.

Source 1

Source path: `results/arithmetic_dimension_bounds_and_effective_high_dimension_search.md`

SHA256: c66ff9573ea560bdabe02b0d4bcb730b3c0110f09d6d3aeff9acfdf9ebd95a3d

Arithmetic dimension bounds and an effective high-dimensional search

Date: 2026-08-12
Status: provenance and reduction theorem; no new numerical nonexistence claim is made.

Search-direction recheck, 2026-10-13. Esselmann's original publisher abstract gives the maximal reflective integral Lorentzian lattice rank as 22, meaning hyperbolic dimension 21. The compact arithmetic case is a different category and remains subject to Vinberg's bound 29. The precise noncompact range, including the exclusion of dimension 20, is restated in Belolipetsky, Section 1, with Borcherds' 21-dimensional example described in Section 7. These are search restrictions, not new results of this repository. The current H22 geometric branch is nonarithmetic if it ever achieves finite volume; its present volume is infinite. No rational-lattice search in the excluded dimensions is warranted.

Current-frontier cross-check. The introduction of Guo–Ma–Zang–Zheng's July 2026 preprint, *Several families of incommensurable noncompact hyperbolic Coxeter polytopes*, still records known compact examples only through dimension 8 and noncompact examples through dimension 21. Its new constructions are families in dimensions 4, 5, 6, 7 and 9; they do not raise either dimension record. This supplies a recent primary-source check rather than relying only on the 2015 survey for the state of known examples.

Dimension 20 is also a target. The known dimensions are not simply every dimension up to 21. Felikson's research overview lists finite-volume examples in dimensions at most 19 and in dimension 21, leaving 20 absent. Thus a nonarithmetic finite-volume H20 example would also fill a new dimension in the sources checked, although it would not raise the maximum-dimension record. Esselmann's arithmetic exclusion in dimension 20 does not prove unrestricted nonexistence there. The existing geometric H19-to-H20 lane should be judged by its specific construction obstructions, not discarded under the arithmetic theorem.

1. The four dimension statements must be kept separate

scope	strongest bound used or immediately available	provenance	sharpness / open range
arbitrary finite-volume hyperbolic Coxeter polytopes	`n <= 995`	Khovanskii–Prokhorov for the noncompact case, together with Vinberg for the compact case	geometry-only; intentionally ignores arithmetic
arbitrary compact hyperbolic Coxeter polytopes	`n <= 29`	Vinberg's 1981/1984 nonexistence theorem	examples are known through `n=8`; for each `n=9,...,29`, the unrestricted existence problem remains open
arithmetic finite-volume reflection groups, compact or noncompact	`n <= 29`	Vinberg's arithmetic nonexistence theorem	no maximality hypothesis is needed

scope	strongest bound used or immediately available	provenance	sharpness / open range
noncompact arithmetic reflection groups in the quadratic/lattice model	`n <= 19` or `n=21`	cusp isotropy gives `k=Q`; Esselmann's theorem then bounds reflective integral Lorentzian lattices	the exceptional dimension `21` is realized by Borcherds, so the numerical ceiling `n<=21` is sharp; `n=20` is impossible

Consequently the displayed `n<=995` in the current AGCat composite theorem is not the best arithmetic bound. It is the bound one gets while importing only a theorem about **all** finite-volume Coxeter polytopes. If Vinberg's arithmetic dimension theorem is admitted, `995` can immediately be replaced by `29`. If the chamber is noncompact, Esselmann improves this further to the disconnected range `n<=19` or `n=21`.

The compact and noncompact arithmetic cases behave differently. For a noncompact arithmetic chamber a cusp supplies a nonzero isotropic vector over the ground field. At every nonidentity real embedding the admissible form is definite, so the ground field must be `Q`. After scaling, one obtains a reflective integral Lorentzian lattice, and Esselmann applies. This argument does not apply to compact chambers: in dimensions at least four they cannot be defined over `Q`, since the rational Lorentzian form would have at least five variables and hence be isotropic by Meyer. For a genuine totally real ground field, the nonidentity definite embeddings defeat the rational-isotropy argument.

The strongest safe statement about the unknown compact range is therefore:

> No compact hyperbolic Coxeter polytope is presently known in dimension at least nine, arithmetic or otherwise, and for each dimension `9,...,29` the unrestricted existence/nonexistence problem remains unresolved. This is not a nonexistence theorem.

There are many conditional or bounded-facet nonexistence results inside this range, but they do not settle arbitrary compact Coxeter polytopes.

2. What the AG/arrangement theorem does and does not contribute

The exact carrier and field funnel make the classification problem more discrete, but they do not presently prove a rank bound. Their contribution is

```
quadric--hyperplane arrangement
      -> integral closed Cartan-cycle coordinates
      -> finite ground-field set
      -> finite bounded-frame representations for fixed rank
      -> finite projective native hosts.
```

The missing arrow for a new dimension theorem is still

```
integral arrangement data -> a rank-independent contradiction above N.
```

Thus a claim that AG has already lowered `29` would be false. The present dimension bounds are geometric/arithmetic imports. The arrangement theorem does, however, turn a proposed improvement into an exact finite computation in each of the finitely many remaining ranks.

3. A rigorous finite reduction for dimensions 22 through 29

Vinberg's refined high-dimensional theorem makes the first target much smaller than the global field theorem alone suggests. For an arithmetic reflection group in dimension `n>=22`, the complete high-dimensional ground-field list is

$$\mathbf{Q}, \qquad \mathbf{Q}(\sqrt{2}), \qquad \mathbf{Q}(\sqrt{5}), \qquad \mathbf{Q}(\cos(2\pi/7)). \tag{3.1}$$

The rational field cannot occur in the **compact** case here. Arithmetic reflection groups are of simplest type; in dimension at least four their defining rational Lorentzian form has at least five variables and is isotropic by Meyer, so the corresponding arithmetic lattice is noncocompact. Thus a compact arithmetic reflection group with n>=22 has ground field in precisely the three-field candidate list

$$\mathbf{Q}(\sqrt{2}), \qquad \mathbf{Q}(\sqrt{5}), \qquad \mathbf{Q}(\cos(2\pi/7)). \tag{3.2}$$

Fix one of these fields and 22<=n<=29, and put m=n+1. Pass from any finite-covolume arithmetic reflection subgroup to its full integral reflection host. Compactness is preserved under this finite-index enlargement. The bounded connected-frame lemma supplies m chamber walls spanning the Lorentzian space and satisfying, for every frame edge,

$$c_{ij} < 62^2 \tag{3.3}$$

at the distinguished embedding. For every simple directed cycle C of length r, including length-two cycles,

$$b_C \in \mathcal{O}_k, \qquad |b_C| < 62^r, \qquad |\sigma(b_C)| \le 2^r \quad (\sigma \neq \sigma_0). \tag{3.4}$$

The Minkowski embedding therefore gives an **explicit finite box** of possible values for every b_C. There are finitely many connected graphs on m vertices. Equality of all simple cycle products determines the ordered reflection representation up to diagonal gauge.

There is a small effectivity point here which must not be hidden behind an assumed invariant lattice. It can be resolved directly from the same cycle data. Write T_i=1-s_i. The trace of a product T_{i_1}\cdots T_{i_r} is the product along its closed directed walk. If it is nonzero, that walk decomposes into simple directed cycles (and loops, for which a_ii=2), so its trace is a product of integral b_C and powers of 2. Expanding words in the s_i therefore shows

$$\operatorname{tr}(w) \in \mathcal{O}_k \quad \text{for every word } w \text{ in the } s_i. \tag{3.5}$$

Burnside's theorem says that the word matrices span A=End_k(V). Enumerate words until m^2 of them, e_1,\ldots,e_{m^2}, form a k-basis, and compute the trace-dual lattice

$$D = \{x \in A : \operatorname{tr}(xe_j) \in \mathcal{O}_k \text{ for every } j\}. \tag{3.6}$$

The trace pairing on A is nondegenerate, so D is a computable full O_k-lattice. Equation (3.5) gives

$$\sum_j \mathcal{O}_k e_j \;\subseteq\; \Lambda := \mathcal{O}_k[s_1, \dots, s_m] \;\subseteq\; D. \tag{3.7}$$

In particular, D/(sum_j O_k e_j) is finite. Starting with the span of 1,s_1,\ldots,s_m,e_1,\ldots,e_{m^2} and repeatedly adjoining all left products by the s_i computes Lambda: the ascending chain stays inside D and must stabilize in that finite quotient. Choose any full seed lattice L_b in V and set L_0=Lambda L_b; this is now a **computed** full projective Lambda-stable lattice, not an oracle input.

Put B=End_Ok(L_0). The annihilator of the finite module B/Lambda computes a nonzero conductor ideal c with cB subset Lambda. Standard exact number-field algorithms compute ideal-class representatives, and the remaining lattices are obtained by enumerating the submodules of the finite modules M/cM that are stable under Lambda. Thus the full-order conductor sandwich really gives a finite algorithm for the stable projective lattices, rather than only a qualitative finiteness statement.

Finally, the effective termination theorem of Belolipetsky–Kapovich turns Vinberg's semi-algorithm into a decision procedure for each resulting arithmetic lattice. Exact cone and vertex-link calculations decide finite volume and compactness.

This proves the following reduction, but not its hoped-for verdict.

Finite high-dimensional decision reduction. The existence and nonexistence of compact arithmetic reflection groups in dimensions `22,...,29`, and the classification of the resulting full integral reflection hosts (which suffices for the reflection-maximal chambers in this reduction), reduce to a finite exact enumeration over the three fields (3.2). A complete empty enumeration would improve the dimension theorem for all arithmetic reflection groups from `n<=29` to the sharp numerical ceiling `n<=21`.

This does **not** assert a finite classification of all nonmaximal reflection subgroups inside a surviving host. Such subgroups are not what the host enumeration or the Belolipetsky–Kapovich termination theorem classifies, and there can be infinite reflection-subgroup constructions. They are irrelevant to the proposed nonexistence decision: every arithmetic reflection subgroup has a full host, and compactness survives the finite-index enlargement.

This reduction is stronger and more concrete than merely saying that the global set is finite. It identifies eight ranks and three fields and gives a certificate format. It is still likely enormous: at `m=30` there are `2^435` labelled support graphs before structural pruning, and the bounds in (3.4) are deliberately crude.

For dimensions `14,...,21`, Vinberg similarly restricts the compact ground field to a short explicit list of quadratic, biquadratic, and real cyclotomic fields. Dimensions `9,...,13` can still be made finite by the global field funnel, but its finite field list must first be materialized rather than left as a Fekete finiteness assertion.

4. Proof-quality computation and certificates

A credible implementation should enumerate constraints rather than raw graphs.

1. **Cycle/frame stage.** Incrementally enumerate connected Cartan supports modulo graph automorphisms and spanning-tree gauge. Enforce exact Coxeter values for intersecting walls, symmetrizability cycle identities, selected signature `(n,1)`, positive definiteness at the other embeddings, and the bounds (3.3)–(3.4). SAT/SMT may propose supports, but every accepted or rejected branch must be checked in exact number-field arithmetic.
2. **Arithmetic-host stage.** Construct the reflection order, enumerate its stable projective lattices through ideal classes and conductor submodules, and quotient by field, lattice, and diagram automorphisms.
3. **Vinberg stage.** Run the effective Vinberg algorithm for each host. Certify root ordering, completeness, exact Gram inertia, all finite and ideal vertex links, and absence of hyperideal extreme rays. Test compactness and deduplicate the resulting full mirror chambers.
4. **Coverage certificate.** Freeze a manifest for every branch with one exact rejection reason: non-Coxeter entry, failed symmetrization, wrong inertia, failed nonselected definiteness, no stable lattice, nonreflectivity, noncompactness, or duplication. A small independent verifier should replay the finite-box coverage and every terminal witness without floating-point tolerances.

For finding examples, run the same pipeline in the opposite order: prioritize small discriminant, small two-cycle values, sparse connected supports, and small conductor. A survivor is then completed by Vinberg's algorithm. This is much more plausible than scanning facet lists directly.

5. Where small ridges help, and where they do not

Bogachev proves that a compact Coxeter polyhedron in `H^3` has a ridge of width less than `5.75`; the same bound holds in higher dimension when the polytope has a three-dimensional face which is itself Coxeter. This improves Nikulin's general width bound `7` dramatically in the affected classifications (the paper reports, in its rank-four examples, reductions from roughly `280` to at most `85` candidates over `Q`, and from roughly `160` to `83` over `Z[sqrt(2)]`).

This is excellent pruning for low-rank classification and for any higher-dimensional branch in which a Coxeter three-face has been certified. It is not currently a universal dimension argument: a face of a Coxeter polytope need not itself be a Coxeter polytope, and no theorem used here forces the required three-face in every compact high-dimensional chamber.

A promising way to obtain a **new numerical** bound is instead to revisit Vinberg's weighted two-face proof. Its combinatorial proposition assigns weights to planar angles and requires **both** a vertex inequality (the total weight at every vertex is at most `cn`) and, for every `k`-gonal two-face, a face inequality (the total weight around it is at least `5-k`). Under those global hypotheses its conclusion has the form

$$n < 8c + 6. \tag{5.1}$$

Vinberg constructs a globally valid assignment with `c=3`, giving `n<30`. Improving this is not merely the calculation of a smaller local maximum. In the high-dimensional arithmetic range one may enumerate candidate local configurations over the restricted fields using the integral cycle coordinates, but one must then exhibit and verify an assignment satisfying the vertex and face inequalities simultaneously for every possible chamber. This can be organized as a finite exact linear-programming/covering certificate only after exhaustive coverage of the configurations has itself been proved. For example, such a global certificate with `c<=23/8` gives `n<29` and therefore rules out dimension `29`. Each further decrease gives a rigorous incremental improvement. This target is potentially far smaller than classifying every full chamber while still retaining an auditable finite certificate.

The obstruction is precise: field finiteness and small ridge bounds alone do not decrease `c`; one must prove exhaustive coverage of all configurations that can occur and a globally valid weight assignment satisfying both families of inequalities.

6. Primary references checked

- E. B. Vinberg, *Absence of crystallographic groups of reflections in Lobachevskii spaces of large dimension*, Functional Analysis and Its Applications **15** (1981), 128–130, https://doi.org/10.1007/BF01082285. This is the source for nonexistence of compact Coxeter and arithmetic reflection groups in dimensions `n>=30`.
- E. B. Vinberg, *Absence of crystallographic groups of reflections in Lobachevskii spaces of large dimension*, Transactions of the Moscow Mathematical Society **47** (1984), 75–112. The expanded paper contains the weighted-face argument and the high-dimensional ground-field restrictions.
- F. Esselmann, *Ueber die maximale Dimension von Lorentz–Gittern mit coendlicher Spiegelungsgruppe*, Journal of Number Theory **61** (1996), 103–144, https://doi.org/10.1006/jnth.1996.0141.
- N. V. Bogachev, *From geometry to arithmetic of compact hyperbolic Coxeter polytopes*, European Journal of Mathematics **8** (2022), 1271–1293, https://arxiv.org/abs/2003.11944.
- M. Belolipetsky and M. Kapovich, *Effective bounds for Vinberg's algorithm for arithmetic hyperbolic lattices*, Sao Paulo Journal of Mathematical Sciences **18** (2024), 558–568, https://arxiv.org/abs/2206.01282.
- N. Bogachev, S. Douba, and J. Raimbault, *Infinitely many commensurability classes of compact Coxeter polyhedra in `H^4` and `H^5`*, Advances in Mathematics **452** (2024), 109826. Its introduction records that dimension eight remains the highest dimension in which a compact Coxeter polytope has been exhibited.

7. Bottom line

The present AGCat proof should be described as computationally enabling, not as already computationally classifying. It makes fields, frame representations, orders, and lattices finite and exact. Vinberg supplies the current compact ceiling and the best high-dimensional field restriction; Esselmann completely settles the noncompact dimension range. The most defensible next theorem target is an exact certified empty search in dimensions `22,...,29`, or an improved local-weight constant in (5.1). Either would be a real new dimension result; merely enumerating more fields would not.

Source 2

Source path: `results/fresh_arithmetic_dimension21_hostile_20261016.md`

SHA256: 29ed1e347545456ee15a0f21e82da8cbb5e2a00ba32bb3897ffd8933df1c9d29

Hostile global-scope audit of the arithmetic bound-21 candidate

No mathematical gap was found in the global transitions audited here. This is a new adversarial source/proof read of the bound-21 composition, its complete $Q(\sqrt{5})$ coverage, and the relevant original geometric and arithmetic statements. It is not a new full attachment-queue or Lannér-catalogue replay. The independent H3 weight reconstruction and prior independent complete queue checks remain explicit dependencies.

The useful new outcome is a separate eight-field obstruction, now with an independent exact review. It reduces a possible **compact dimension-21** chamber to the fields $Q(\sqrt{5})$ and $Q(\sqrt{2},\sqrt{5})$, without applying the dimension-22 three-field restriction below its range. It does not exclude either remaining branch.

Global failure modes checked

Compactness and simplicity. The proof uses a compact chamber throughout its face-weight argument. Its Klein-model image is an ordinary bounded convex Euclidean polytope, so the simple-polytope h-vector theorem applies. Compact acute Coxeter chambers are simple; their finite corners have positive-definite Gram matrices. The corresponding whole-chamber diagram contains no affine principal subset. These are the hypotheses in Vinberg, §3.1–3.2 and Propositions 7–9. They cannot be inferred merely from one compact face of a noncompact chamber. The earlier compact-face counterexample therefore does not contradict this proof, but would invalidate extending this argument unchanged to general noncompact polytopes.

The actual-face step is also valid. For a positive-definite principal Z-Gram B, let H be a strict interior timelike vector and put $b=B^{-1}(H\cdot n_i)$. Entrywise nonnegativity of B^{-1} gives $b\le 0$. The orthogonal projection $H_o=H-\Sigma b_i\ n_i$ remains timelike and makes every other wall pairing strictly more negative or unchanged. Thus a positive-definite subset really gives an intersection of facets. A positive affine kernel instead gives a nonzero future null vector satisfying every chamber inequality, contradicting compactness. Opposite extensions of a quadrilateral are therefore genuinely indefinite, not merely non-positive-definite numerical matrices. The argument concerns the ambient chamber; it never assumes that a two-face or facet is itself Coxeter.

A stronger rank bridge. The queue's corank-one requirement can be justified without a geometric drawing of the projected polygon. A k-node quadrilateral core has a positive-definite corner on $k-2$ nodes, and its opposite minimal Lannér subdiagram supplies a negative direction. Therefore its positive index is at least $k-2$, its negative index is one, and its nullity is at most one. For the complete star in dimension n, $k=n+2$ while the ambient Lorentz space has dimension $n+1$. Its nullity is consequently **exactly one**. This proves both the complete-star singularity requirement and the dichotomy between nonsingular proper cores and singular stopping choices.

This also explains why a sealed nonsingular core cannot be repaired by white spectators. If every remaining white root is orthogonal to it, the extra block is positive definite. The direct sum stays nonsingular. If a remaining component touches it, there is a first touching white node, and that node must occur in the complete one-white domain. No long path can avoid its own first contact.

Kernel and stopping-state completeness. A principal Lorentz span containing a timelike vector is nondegenerate as a subspace of the ambient Lorentz space. Hence a kernel vector of its Gram matrix is a literal linear relation among its normals. Every later root must respect that relation; a nonnegative relation is impossible because pairing it with a strict interior point is negative. These facts justify the relation/sign gates, rather than treating Gram kernels as harmless numerical degeneracies.

The critical potential omission would be retaining only final sealed singular states. That would lose an actual full star that stops at a singular state which also admits further touching extensions. The completed finite and dotted queues retain **every** singular zero-touch choice, including expanded states and boundary states. The binder directly reevaluates all 2,661 earlier and queue obligations with the new rule. It does not inherit their old numerical payments. The independent final finite replay covers all 5,172 states and 2,070 singular obligations; the dotted queue and its exact arithmetic parameter domain remain separately reviewed dependencies.

Lower-dimensional reuse of the dimension-29 queue. A full n-dimensional star has n+2 nodes. Since the original published compact bound gives $n \le 29$ before the search, 31 nodes is a legitimate upper bound. The same complete queue can serve dimension 22: an actual 24-node star is one of its singular stopping choices, and all larger rows are simply infeasible at that dimension. Node counts only increase along attachments. No claim that a nonsingular 24-node core could be a complete dimension-22 star is made. The newer binder recomputes spectator rank as $24-k$; it does not reuse the original rank-28/29 spectator metadata.

Minimal unions and dotted distances. The use of no-short-path cores is justified by an unchanged nonnegative baseline which already pays any short marked corner. The relevant minimal unions include shared whites and every cross contact; they are not assumed to be complete stars. For the surviving dotted domains, the common Lorentz block already uses the unique negative direction, so its endpoint Schur complement is positive semidefinite in every possible extension. This produces a uniform arithmetic bound on the dotted cyclic invariant. The lattice enumeration is thus a finite exact consequence of arithmetic integrality and both real embeddings, not a scan of sampled distances. The full parameter/attachment reviews, rather than the small bound-21 binder alone, are necessary for this premise.

Whole-forest and spectator quantifiers

For the dimension-22 rule, a,b,c count **entire** A1, $I_2(5)$,H3 components. The capped values select coefficients only. Actual denominators are a,2b,3c. A1 pairs may target $I_2(5)$ and H3; $I_2(5)$ pairs may target H3; new H3 pairs exclude all three hub types as destinations. This partitions the pair families, avoiding both overlap and missing incoming payments. At fixed exact counts the multiplicities cancel these denominators, so the component dynamic program genuinely maximizes the whole vertex sum.

At a zero-touch k-node core the remaining white block is the same at every corner. The binder enumerates every common triple e,f,g with $e+2f+3g \le 24-k$ and residual rank zero or at least two. A2 and A3 realize every such residual rank, and neither introduces an uncounted hub. Internal hub counts may differ among the four corners; the same external triple is added separately to each. It would be invalid to optimize or choose spectators independently at each corner, or to put capped values in the denominators. Neither happens here.

The independent H3 review reconstructs the literal graph weights, finite-component models and forest dynamic program without producer or optimizer imports. The complete binder extends those payments to every older branch, accounting for 1,634 dimension-feasible rows and 91,726 common spectator triples. Nonnegative bonuses preserve the previous triangle and persistent short/H4 payments. No H3 bonus was retroactively used as a pruning condition in a growing attachment core.

Strict counting and field transfer

The planar angles at a vertex correspond to unordered pairs of its n incident facets. Triangle and quadrilateral payments give total weight at least $5f_2 - \mathrm{binom}(n,2)f_0$; k-gons with $k \ge 5$ need no additional structural assumption because their required value $5-k$ is nonpositive. For n=22, paired h-vector coefficients have minimum 55. The endpoint terms $h_0 = h_{22} = 1$ have strictly larger coefficients, so $f_2/f_0 > 55$. The average weight is strictly greater than 44, contradicting the exact maximum 21977/500. Strictness does not require an unproved lower bound on every middle h-entry.

The composition only invokes Vinberg's three compact fields in dimensions **22–29**. This exact boundary is also explicit in Nikulin's primary account, introduction and §4.1. $Q(\sqrt{2})$ and K_7 use their full independently reviewed external-attachment/rank arguments, not just a list of small admissible matrices. The separate noncompact arithmetic bound is not derived from this compact h-vector proof. Arithmetic reflection lattices are of simplest type, so the composition has not omitted an additional type of arithmetic reflection lattice. See Belolipetsky, Lemma 2.1 and §7.

The final small scope checker separately checks the complete field/dimension Cartesian product and literal obligation identities. Its successful sums cannot by themselves establish the geometric premises above; conversely, I found no missing

global premise after checking those transitions explicitly.

What can actually happen next at compact dimension 21

The larger compact field list beginning at dimension 14 has twelve fields. The new distinguished-bond proof excludes eight of them in dimensions at least 21; the existing $Q(\sqrt{2})$ and K_7 candidates exclude two more. Thus only $Q(\sqrt{5})$ and $F=Q(\sqrt{2},\sqrt{5})$ remain at dimension 21. This is a consequence of the new field theorem and its separate independent review, not of extending the three-field theorem outside its range.

There is a precise simplification for F. Nikulin, §3.1, Figure 1 and formula (4) explicitly gives the ground fields of all arithmetic Lannér diagrams of order at least four as Q,$Q(\sqrt{2})$,$Q(\sqrt{5})$. Since a principal Lannér diagram in an arithmetic chamber has the same ground field, **F has only order-two/three Lannér subdiagrams**. This step uses the published arithmetic classification; mere containment of all low-label entries in F would not prove it.

The triangle difficulty is real, not an absent computer case. The cited complete arithmetic list contains (2,5,8) and (4,5,5), both with full ground field F. For (2,5,8), the squared contacts generate $\sqrt{5}$ and $\sqrt{2}$; its 8 bond has only one nonright attachment. For (4,5,5), the squared 5 contact supplies $\sqrt{5}$ and the cyclic product supplies $\sqrt{2}$. These defeat the new distinguished-bond premise. A complete next F argument must handle their lower-label geometry, dotted opposite pairs and arbitrary white extensions. It cannot infer dimension$\leq$20 merely from the eight-field result.

Validation and literature limits

This audit is a source-level mathematical challenge, not a second run of every large catalogue. I found no counterexample or unclosed global quantifier in the stated bound-21 candidate. It remains an internally reviewed computer-assisted research proof, not formal verification or external peer review.

A bounded primary-source comparison included the 2026 Mühlherr–Weiss publication, §7, which still reports the classical published high-dimensional exclusions and the known dimension-21 example. This is not an exhaustive literature-absence or novelty certificate. The example is noncompact; it prevents lowering the all-arithmetic bound below 21 but does not prevent a compact-only improvement.

The earlier finite-queue independent generation and fresh saved replay remain confirmed: session 23308 completed successfully with 5,172 states, 2,070 obligations and no pending frontier. The old dimension-27 prose's broader claim about a fresh replay of the *fast producer* is historical imprecision already clarified in the previous fresh audit. The producer had generation plus structural database/map checks; the independent reviewer had full generation and fresh replay. No frozen file has been changed.

Frozen inputs directly read in this audit

File	SHA-256
Final composition source	1b7fa8cb98cd0c1f48a54b2b8d940ba4a6143a5f35ac360b6b0edea4123b5ceb
Final composition certificate	451d04e5305c1bf15148e83aa2c35292f78686b9816f81dec73f19af415e80ab
Coverage source	661585192d0e7f21add453336604905c04509872cad9bd9c66299f58a69f2c52
Coverage certificate	dcf4be8a774f5f1961df2b3d882bd79a4d9a22c0e236259c26829882a603c9f1
Independent H3 source	1d466f25f59fd3df1f0059456bdd464060c85430eb9784d6eff3fa31234dc19a
Independent H3 certificate	576418cd7c12bdcca6896d66e91a73fcabe3578621c4f73d71ae6420e2f0aaa1

All mathematical statements above keep their stated compactness, ground-field and reviewed-computation dependencies. The new eight-field package is separate from the frozen bound-21 composition.

Source 3

Source path: `results/fresh_arithmetic_sqrt2_local_20261015.md`

SHA256: 56903969d6add015c3b4aa1de8729a55624fd1b566cef0be62c067b626b3135c

Arithmetic triangular faces at E7 and E8 over $Q(\sqrt{2})$

The complete finite-contact one-root census leaves triangular completions at **10 of the 21 marked pairs in E7** and **9 of the 28 in E8**, for the arithmetic ground field $Q(\sqrt{2})$. The exact checker and data use rational arithmetic and exact quadratic-field signs. These are local restrictions, not a dimension theorem; quadrilateral completions remain open.

Number the E_r diagram by the path `0–1–…–(r–2)` with the extra edge `2–(r–1)`. All old bonds have label3 and all root squares are2. The allowed marked pairs are:

Component	Pairs admitting a triangular local star
E7	01, 02, 06, 12, 16, 23, 26, 34, 36, 45
E8	01, 02, 07, 12, 17, 27, 34, 45, 56

For example, `06` denotes the unordered pair of nodes0 and6, not a bond label. All pairs absent from the table are excluded as triangular two-faces in the stated arithmetic setting. The argument allowing other orthogonal components of a full vertex diagram is given below. No assertion is made that the remaining local stars fit together in one compact polytope.

Complete finite alphabet and squareclasses

Vinberg's arithmetic criterion places squared normalized pair products and all cyclic products in the ground field, with conjugate Gram matrices positive semidefinite. Nikulin, §2, recalling Vinberg's criterion.

For a finite angle π/m, `4cos²(π/m)=2+2cos(2π/m)` belongs to $Q(\sqrt{2})$. The real cyclotomic degree therefore implies `φ(m)≤4` for m≥3. The complete candidates are m=3,4,5,6,8,10,12. Direct evaluation removes5,10,12; including right angles leaves **2,3,4,6,8**. This is a field restriction, not a search cutoff.

Let p be the nonnegative contact row of the new root against the connected old E diagram. For any two nonzero contacts p_i,p_j, the old path joining i,j has products−1. Its cycle through the new root implies `p_i p_j∈Q(√2)`. Hence all nonzero contacts lie in one of three classes:

- mixtures of 1 and $\sqrt{2}$, corresponding to labels3 and4;
- only $\sqrt{3}$, corresponding to label6;
- only $\sqrt{(2+\sqrt{2})}$, corresponding to label8.

No mixed classes are missed. The first exceptional radical $\sqrt{3}$ is not in $Q(\sqrt{2})$; $\sqrt{6}$ is not either. The field norms of `2+√2`, `2(2+√2)` and `3(2+√2)` are2,8,18, respectively, so none is a square in $Q(\sqrt{2})$. These facts exclude every remaining cross-class product. Zero contacts are allowed everywhere.

The script enumerates all `3^r+2·2^r` profiles for each r=7,8, including duplicated zero rows. No dotted contact can occur in a triangular star: its new side shares a finite corner with each old black side, and every old white wall contains the face.

Exact Schur tests and local realization

Let G be the old positive-definite Gram and J its rational inverse. The full one-root extension is Lorentzian exactly when

$q = p^{\top} J p > 2.$

Its nonidentity conjugate is positive definite exactly when $q' < 2$. Nondegeneracy makes strict positivity necessary even if one starts with the semidefinite version of the arithmetic criterion. For the label8 class, the two extensions of the field embedding differ by the sign of the new root and have the same Schur value.

Replacing old node i by the new root gives a finite vertex precisely when

$2 - q + (Jp)_i^2 / J_{ii} > 0.$

Every pair of such replacement nodes is a possible marked pair of a triangular star. The original and the two replaced diagrams are positive definite; eliminating the white support gives a three-wall Lorentz Gram whose two-wall principal blocks are positive definite and off-diagonal entries nonpositive. Its three vertex rays are timelike in one cone and feasible, so they bound a compact triangle. As in the frozen local-face construction, a small inward white perturbation realizes the ambient local chamber. Conjugate positive definiteness rules out affine principal subdiagrams.

Component	All profiles	Lorentz and conjugate-positive	Profiles with two or more finite replacements	Marked pairs covered
E7	2443	19	10	10
E8	7073	13	7	9

The label6 class contributes no survivor. Label8 contributes two E7 profiles and one E8 profile, all with at least two finite replacements. The remaining survivors use labels3 and4. Cyclic products of these local stars are algebraic integers; the label8 star has only one nonzero contact and its squared product is $2+\sqrt{2}$. These local arithmetic checks do not establish finite volume of the ambient chamber.

Other components of the vertex diagram

Suppose the full vertex diagram is E⊕Z, the marked pair lies in E, and x closes a triangular face. The subsets obtained by deleting either marked E node and adding x are spherical. If x is right to E, the full indefinite star would already be indefinite on Z+x, which is contained in both finite corners: impossible.

If the E+x Schur cost exceeds2, the census applies. Its two replacement tests are necessary because passing to a principal submatrix preserves positive definiteness. Its conjugate is positive definite because its determinant is nonzero and it is a principal submatrix of the conjugate ambient Gram. Extra components can only restrict the listed possibilities.

The exact inverse of E8 has minimum diagonal2 and strictly positive entries. Thus every nonzero contact row has cost at least2. Equality is an affine obstruction, excluded by compactness. For E7, the only nonzero profile with cost below2 is a single label3 contact at node5, with cost3/2. Multiple contacts already cost at least3 before their positive cross terms, and every other singleton has cost at least2.

In that E7 exception, x has at most one neighbor in any connected component of Z, since a second would produce a cycle in a finite corner. A corner retaining the E contact connects x to at least three nodes in every attached component. Such a finite connected Coxeter diagram cannot contain a label6 or8 bond. All its bonds therefore have label3 or4, whose squares are rational. Eliminating these finite trees contributes a rational Schur cost. Unattached components can be discarded. The total Schur cost is consequently rational and exceeds2, so its negative Schur complement persists at the other field embedding, contradicting arithmeticity. The affine cost-equality case is excluded as above.

This covers the potential loophole in which E+x alone is positive definite but x's contacts to other vertex components make the full star indefinite. It does not classify quadrilateral stars or combine different triangular stars globally.

Status

Generation and saved replay pass. Independent review is recorded separately when complete. The script source and data are frozen; any stronger search uses a new namespace. The earlier unrestricted local-face witnesses remain valid in their stated nonarithmetic scope and are unchanged.

```
python scripts/fresh_arithmetic_sqrt2_local_20261015.py --verify
```

Source 4

Source path: `results/fresh_arithmetic_sqrt2_quad_rank_20261015.md`

SHA256: 5bf3238e677b3320c686b1fa5fbe268fc8890556bbb1e0d598c05195678d0873

A restricted quadrilateral-star obstruction over $\mathbb{Q}(\sqrt{2})$

The exact checker and saved certificate exclude the complete-star topology specified below. This is a local arithmetic obstruction, not a classification of all quadrilateral stars and not a new global dimension bound. The independent review reconstructs the determinant identities and every bounded placement with separate Fraction arithmetic.

Theorem scope. Consider an actual quadrilateral two-face of a compact arithmetic Coxeter polytope with ground field $\mathbb{Q}(\sqrt{2})$. Its white nodes are the full spherical support of the face, and its four black nodes are its four sides. Suppose its affected connected diagram consists of two disjoint chordless cycles of orders 3, 4 or 5, each containing one opposite black pair. Each cycle has exactly one bond 4 and all its other bonds 3. At least one cycle has order 4 or 5. Suppose the cycles are joined by a single white bridge, allowing arbitrary pendant spherical white trees, each attached at precisely one white cycle node. Other white components may be orthogonal. There are no other contacts. Then this cannot be the complete star of that quadrilateral.

A unique connector path of two or more bonds is also impossible: the two original cycle subdiagrams are orthogonal Lannér diagrams and already give two negative directions.

The direct-bridge theorem allows any number and size of the specified pendant trees; this part is proved analytically, not by sampling trees. It does **not** allow shared white nodes, extra black contacts, further cross edges, or both cycles of order 3. These exclusions from scope matter.

Exact finite placement check

Fixing the distinguished 4-bond on each cycle still covers every labelled placement of its two black nodes and white bridge endpoint. Both possible bridge labels, 3 and 4, are tested. The census contains **4,032 labelled placements**, not 4,032 isomorphism classes.

Ordered cycle orders	Placements	All four corner diagrams positive definite
3,4	72	6
4,3	72	6
3,5	180	8
5,3	180	8
4,4	288	0
4,5	720	0
5,4	720	0
5,5	1,800	0
Total	**4,032**	**28**

All 3,528 placements with both cycle orders at least four already fail a finite-corner test. The other 28 survivors have nonzero full Gram determinant. The argument below also excludes all their allowed pendant-tree extensions. Some survivors have long marked paths, so the full-star rank condition supplies information beyond the four separate corner tests.

The primary uses rational tree elimination for the finite-corner checks and symbolic determinant identities. The independent review uses full-matrix quadratic-field LDL and Gaussian elimination, and derives the cycle formula from determinant matchings. Neither finite-placement census is presented as an exhaustion of arithmetic quadrilateral stars.

Proof with arbitrary pendant trees

Use unit-square normal vectors. The finite-contact alphabet over this field is $2, 3, 4, 6, 8$, as derived in the frozen quadratic local package and its independent review. A bridge or pendant bond belongs to an actual spherical corner component of rank at least three. Finite labels 6 and 8 must be isolated rank-two components, so every such bond here has label 3 or 4. In particular its squared Gram entry is rational. This step is needed: a label-8 square would otherwise introduce a new irrational term in the bridge correction.

Schur-eliminate the pendant white trees. Their positive-definite Gram matrices are forest matrices with rational squared bonds. Successive leaf elimination therefore subtracts rational, nonnegative quantities from the diagonal of the unique attachment node and creates no additional edge. The resulting cycle diagonals p_i are positive rational numbers at most one. Positivity follows from the original positive-definite white support. Orthogonal white components can be removed as positive factors.

For a cycle of order $r \in \{3, 4, 5\}$, its reduced determinant has the form

$$\det L = a(p) - b_r\sqrt{2}, \qquad b_r = 2^{1-r} > 0.$$

The two oriented full-cycle terms are the only irrational determinant terms; all matching terms use rational bond squares. Thus the coefficient b_r is independent of the diagonals. Every proper principal minor is a forest determinant and is rational. Direct calculation gives

$$a(1, \dots, 1) = 0 \quad (r = 3), \qquad a(1, \dots, 1) = -\frac{1}{16} \quad (r = 4, 5).$$

At the other arithmetic embedding the full Gram is positive semidefinite. This property survives Schur elimination of the white trees. The conjugate cycle block is actually positive definite: its determinant is $a(p) + b_r\sqrt{2}$, which cannot vanish because $a(p)$ is rational and $b_r \neq 0$. Increasing the reduced diagonals to one preserves positive definiteness. The derivative of the determinant with respect to any diagonal is a positive proper principal cofactor, which is rational and unchanged by conjugation. Consequently

$$a(p) \leq 0 \quad (r = 3), \qquad a(p) \leq -\frac{1}{16} \quad (r = 4, 5).$$

Write the two reduced cycle determinants as $a - b\sqrt{2}$ and $A - B\sqrt{2}$. A single bridge of magnitude δ, joining nodes u, v, gives

$$\det G = (a - b\sqrt{2})(A - B\sqrt{2}) - \delta^2 \det(L - u) \det(M - v).$$

The second term is rational. The coefficient of $\sqrt{2}$ is therefore

$$-aB - Ab > 0,$$

because at least one cycle has order four or five. Thus the reduced determinant is nonzero, and restoring the eliminated positive-definite factors does not change that conclusion.

But the **complete** star has $n - 2$ independent white nodes and four black nodes in a Lorentz space of dimension $n + 1$. Its Gram determinant must vanish. This contradicts the nonvanishing just proved. Applying this rank count to a smaller subdiagram while omitting other connected white nodes would be invalid; completeness of the specified topology is an explicit hypothesis.

Configurations deliberately left open

The checker verifies two singular examples outside the theorem:

- The six-node path with labels $8, 3, 4, 3, 8$ has determinant zero. Its long marked corner is of type F4.
- Two bare $(3, 3, 4)$ cycles joined through their white vertices by a 4-bond have determinant zero and four positive-definite F4 corners. Here both rational determinant parts are zero, so the strict coefficient argument does not apply.

These are local diagrams, not certificates of a complete compact polytope. They suggest testing the weight rule "distance at most two, plus one extra unit on the end pair of an entire F4 component." The component bound would remain $2r-2$, but no exhaustive quadrilateral inequality for this rule has been proved here. Shared-white configurations, other Lannér triangles, dotted opposite pairs and additional contacts remain separate tasks.

A further checked follow-up connects the dotted-pair and label-8 cases. Take a dotted pair a, y, a cycle (b, u, v, w, d, b) with its only 4-bond on b, d, and one extra bond a, v of label 3. All unlisted pairings are right. Give node a reduced diagonal p, all other diagonals one, and write its pairing with y as $-x$. Direct symbolic elimination gives

$$\det G = \frac{1+\sqrt{2}}{64}\bigl(4x^2 - 4p + 1 - \sqrt{2}\bigr).$$

Attaching a white A_ℓ tail by a 3-bond at the **black** node a gives $p = (\ell+2)/(2(\ell+1))$. Hence singularity forces the norm-two squared contact

$$D = 4x^2 = 1 + \frac{2}{\ell+1} + \sqrt{2}.$$

For $\ell = 0$, $D = 3+\sqrt{2} > 4$, genuinely dotted. For $\ell = 1$, $D = 2+\sqrt{2} < 4$, exactly label 8; the minimal indefinite piece changes to a $(8, 3, 2)$ triangle. For $\ell = 2, 3$, the forced values $5/3+\sqrt{2}$ and $3/2+\sqrt{2}$ are not in the allowed finite-contact alphabet and are not algebraic integers. This identity was checked separately with a symbolic seven-by-seven determinant; it is a follow-up lead, not part of the complete placement census or a global exclusion. In particular, attachment at a black node is outside the pendant-white theorem above.

Reproduction and status

```
python3 scripts/fresh_arithmetic_sqrt2_quad_rank_20261015.py
python3 scripts/fresh_arithmetic_sqrt2_quad_rank_20261015.py --verify
python3 scripts/fresh_arithmetic_sqrt2_quad_rank_20261015_review.py --verify
```

Primary generation and saved replay pass. The independent generation, saved replay, source bindings and syntax checks also pass. Source manifests bind the preceding quadratic local proof and its independent review. This package is frozen; no prior frozen artifact is modified.

Source 5

Source path: `results/fresh_compact21_single_field_20261016.md`

SHA256: 0988994cfe0c29943efdea7613afb299eb3783a3c2e80da73719a29465d972eb

Compact arithmetic dimension21: one field remains

Internally reviewed research reduction: if a compact arithmetic Coxeter polytope of dimension21 exists, its ground field must be $\mathbf{Q(\sqrt{5})}$. This does not exclude dimension21 itself or lower the overall arithmetic bound21 candidate. External validation and priority remain open.

The binder and certificate combine the frozen two-field reduction with the new F25 proof and its independent review. Generation and a fresh saved replay pass. All recursive source manifests are checked; this composition does not rerun inherited large enumerations.

The thirteen dimension21 branches are retained explicitly: the rational field is excluded by the published isotropy argument; eleven nonrational fields now have reviewed bound20 candidates; $Q(\sqrt{5})$ remains open. The larger dimension14 field list is used here, never the three-field restriction valid only from dimension22. The new exclusion concerns exact field $Q(\sqrt{2},\sqrt{5})$, and is not applied to its proper subfield $Q(\sqrt{5})$.

The arithmetic bound21 candidate, the published noncompact arithmetic bound21, the unrestricted compact bound29, and the unrestricted finite-volume 163 candidate are unchanged. No existence result follows from the remaining field branch.

```
python3 scripts/fresh_compact21_single_field_20261016.py
python3 scripts/fresh_compact21_single_field_20261016.py --verify
```

Source 6

Source path: `results/fresh_compact_global_bound_20261015.md`

SHA256: 0f7d1a4516a011c52ac6106721f820802d47b2cbd7a36840b20506be961dc59a

A corrected small-face weighting for the cubic-seven arithmetic field

Independently reviewed result: a compact arithmetic hyperbolic Coxeter polytope with ground field **Q(cos(2π/7)) has dimension at most20**. The argument below uses a simultaneous quadrilateral-star rank constraint, rather than excluding individual marked spherical angles. It does not improve the unconditional compact bound29, and it makes no novelty or priority claim.

The exact checker and certificate verify43,836 relaxed local graph cases, both symbolic determinant identities, their cubic-field remainders,101 finite-diagram graph records, the exceptional four corners, and the final h-vector arithmetic. The geometric reduction and all-rank arguments are written here; the finite computations do not replace them.

A separate independent review passes generation and saved replay. Its standalone Fraction polynomial engine reconstructs the determinant identities and remainders, an independent35,904-graph enumeration leaves the same nine supports, and122 finite graphs and all four corner matrices are checked. The hostile proof review separately checks arbitrary extra white contacts, rational branch elimination, the full-star rank condition and the assertion that the exceptional D_5 components are entire. No producer arithmetic is imported.

The separate arithmetic field package supplies the complete cubic-seven finite-label/Lannér restrictions and the weaker dimension28 fallback. Its source/data are unchanged. The present package does not alter any frozen construction or search cursor.

Inputs and normalization

Normalize facet normals to square1, so a finite bond labelled m has product $-\cos(\pi/m)$. We use the standard compact Coxeter facts: vertex diagrams are spherical forests; elliptic subsets correspond to intersecting facets; affine subdiagrams are absent; a minimal indefinite subdiagram is Lannér. We also use arithmetic cyclic products and the symmetric nonnegative h-vector. These are the published inputs, not new computational conclusions. Vinberg, §§1–3 and7, especially Proposition17.

For this ground field the finite labels are2,3,4,6,7,14. Lannér diagrams have order at most3, and every Lannér triangle has a7 or14 bond. The exact field package separately checks the seven allowed triangle types. In a finite diagram any bond of label at least6 is an isolated rank-two component: adding a nonright adjacent bond gives a three-node path determinant at most$1-3/4-1/4=0$.

Set $c=\cos(\pi/7)$, $t=1-c^2$. Both c and c^2 have degree3 overQ, with irreducible polynomials

$$P(c) = 8c^3 - 4c^2 - 4c + 1 = 0, \qquad 64(c^2)^3 - 80(c^2)^2 + 24c^2 - 1 = 0.$$

We have $c>9/10$. Indeed $c>\cos(\pi/6)>4/5$, P is strictly increasing on$[4/5,1]$, and $P(9/10)=-1/125$. No upper bound on an unexamined integer label is being imposed: the complete label list is an arithmetic field consequence.

A quadrilateral with no short marked corner

Let W be the spherical white support of an actual compact quadrilateral two-face, and let a,b,c_o,d be its four black sides in cyclic order. The symbol c_o denotes a node; c continues to denote $\cos(\pi/7)$. Each W plus an adjacent black pair is

positive definite. The opposite extensions W+a+c₀ and W+b+d are indefinite. They cannot be positive definite by the face correspondence, and cannot be affine in a compact chamber. Their minimal indefinite subsets L,M include both respective black nodes; hence they have order2 or3.

Assume no adjacent black pair is connected by a path of length at most2 in its finite corner diagram. The following reduction is exhaustive.

1. All adjacent black pairs are right. A Lannér2 opposite pair would be orthogonal to the other minimal indefinite diagram: any cross black bond gives a length-one marked path, while a cross black-to-white bond gives a length-two marked path. Two orthogonal indefinite diagrams violate Lorentz index one. Thus both L and M are triangles.
2. Their white nodes u,v are distinct. Their only possible connecting bond is u–v. A high bond involving a white node would be isolated in each finite corner and would force u–v right, again making L and M orthogonal. Therefore the high bonds are the two opposite black pairs a–c₀ and b–d, and u–v is nonright.

The checker enumerates every binary support of the two connected diagrams, all cross-bond subsets, shared or separate white nodes, and all choices of distinguished high bond. It only imposes the necessary finite-corner forest/isolation conditions and absence of a short marked path. The nine remaining supports are precisely the nonempty attachment patterns on either side of the two high opposite bonds, joined by u–v. This is a relaxed complete finite verification, not a numerical Gram test.

Eliminate label14 and dense attachments

The two opposite I_2 blocks have a positive-definite direct sum. Eliminate them by a Schur complement. A single$\pi/3$ contact to an $I_2(7)$ block costs more than

$$\frac{1}{4(1-(9/10)^2)} = \frac{25}{19}.$$

A single$\pi/3$ contact to $I_2(14)$, or two$\pi/3$ contacts to both roots of $I_2(7)$, costs more than

$$\frac{1}{2(1-c)} > 5.$$

Larger contacts only increase these costs, since the inverse I_2 Gram has positive entries. If either opposite bond is14, or either white node touches both endpoints of its opposite7 bond, the two residual white diagonals are less than-4 and$-6/19$. Their off-diagonal remains their old product: the two white nodes have no cross contacts to the other black block. Its square is at most1. Thus their determinant exceeds

$$4\frac{6}{19} - 1 = \frac{5}{19} > 0.$$

This supplies a negative-definite two-plane, impossible in a Lorentz Gram. Consequently both opposite bonds are7, and every white neighbor touches exactly one endpoint of its associated opposite pair.

Let U consist of all white neighbors of a or c₀, and V of all white neighbors of b or d. They are nonempty and disjoint. Each triple consisting of an opposite7 pair and one of its white neighbors is Lannér. Applying the preceding nonorthogonality argument to every such pair of triples forces every U–V bond nonright. Since W is a forest, min(|U|,|V|)=1. Interchange the two pairs if necessary and write U={u}.

Each of b,d has at most one white neighbor: two neighbors v,w would give the cycle b–v–u–w–b, or its d analogue, inside the finite diagram W+b. Thus |V| is1 or2. These statements cover all white attachments, not merely the initially selected L,M.

Every bond touching the resulting white core has label3 or4. A label6 or higher next to another nonright bond violates spherical isolation, and label5 is excluded by the ground field. Any remaining white component attaches to this core at at most one node, because W is a forest. Eliminate all such branches and discard orthogonal white components. Leaf elimination has rational diagonal corrections: every squared branch bond is rational, and the branch is a finite tree. The surviving white diagonals p,q,r are rational, positive and at most1. Any nonempty attached branch decreases its attachment diagonal strictly.

The full star has rank rank(W)+3: its black projections span the Lorentz3 normal space of the compact quadrilateral. Thus the reduced six- or seven-node Gram is singular. Merely checking a local Lorentz Gram without this rank requirement would be insufficient.

The rank condition leaves exactly one exception

One white neighbor on each side

Write the two surviving white diagonals p,q, their bond magnitudeδ, and their contacts to black endpoints α,β. Put A=α²,B=β²,dₒ=δ². The full six-node determinant is

$$(pq - d_0)t^2 - (pB + qA)t + AB.$$

All coefficients are rational and pq−dₒ>0 because the reduced white Gram is positive definite. This is a nonzero polynomial of degree2 in c², so cannot vanish. This excludes, in particular, the apparent six-node path7–3–3–3–7 as an actual quadrilateral star. That path can have Lorentz signature; its excessive rank prevents it from being the required two-face star.

Two white neighbors on one side

Name the white nodes v,u,w along a path, with cₒ–u of magnitudeα, b–v of magnitudeβ, and d–w of magnitudeγ. Write δ,ε for u–v,u–w. After branch elimination the white Gram is

$$H = \begin{pmatrix} p & -\delta & -\varepsilon \\ -\delta & q & 0 \\ -\varepsilon & 0 & r \end{pmatrix} > 0.$$

Use A=α²,B=β²,C=γ²,dₒ=δ²,eₒ=ε² and L=δεβγ. The full seven-node determinant is

$$N(c) = (pt - A)(qrt - qC - rB + BC) - d_0(rt - C)t - e_0(qt - B)t - 2Lct.$$

This identity is checked directly from the literal7×7 matrix. All its coefficients are rational. The only point needing arithmetic beyond squared bonds is L: the cycle b–v–u–w–d–b has cyclic product a nonzero rational multiple of Lc, which belongs to the ground field. Since c belongs to that field and L belongs toQ($\sqrt{2}$), the cubic field's absence of a quadratic subfield forces L rational.

The coefficient of c in the remainder of N modulo P is

$$\frac{pqr - d_0 r - e_0 q - 8L}{8} = \frac{\det H - 8L}{8}.$$

Singularity therefore implies detH=8L. On the other hand, positive-definite determinant monotonicity under increasing diagonal entries gives

$$\det H \le 1 - d_0 - e_0 \le \frac{1}{2} \le 8L.$$

The last inequality uses δ,ε,β,γ≥1/2. Equality throughout forces p=q=r=1 and δ=ε=β=γ=1/2. The equality of white diagonals removes every off-core white branch. Substitution into the remaining remainder gives

$$\frac{(4A - 1)(16c^2 - 9)}{64} = 0,$$

so α=1/2 as well. The entire affected diagram is now fixed: white Aₛ, both opposite bonds7, and all five other bonds3. Its determinant factors as

$$N(c) = \frac{(4c+3)(8c^3-4c^2-4c+1)}{64} = 0.$$

This is an exception to the plain distance-two rule, not a contradiction. Two corners have type A_4+A_1 and the other two have type D_5. At each D_5 corner the marked nodes are the far leaf of the length-two arm and one of the two short leaves, at distance3. Because the white diagonals stayed1 and all other black–white contacts were exhausted, these D_5 diagrams are **entire connected components** of the corresponding full vertex forests. Arbitrary other white components are orthogonal to them.

Corrected weights and the global contradiction

Assign a planar-angle weight from its finite vertex diagram as follows:

- weight1 if its two marked nodes have graph distance at most2;
- one additional unit if they form an isolated $I_2(7)$ or $I_2(14)$ component;
- weight1/2 if they are the far leaf of the length-two arm and a short leaf in an entire D_5 component;
- weight0 otherwise.

The last rule concerns exactly two pairs in each D_5 component; they have distance3 and hence receive no base unit.

A triangular two-face has its three black nodes forming a Lannér triangle: every proper black subset with W is spherical, and the minimal indefinite subset has order at most3. It therefore has a high7/14 bond. The corresponding corner has that bond as an isolated I_2 component and receives weight2. A quadrilateral either has a short marked corner of weight1, or is the classified exception and receives1/2+1/2 from its D_5 corners. A face with $k \geq 5$ sides has nonnegative weight, which is at least$5-k$. Thus every two-face has total weight at least$5-k$.

For a connected finite diagram of rank r, the distance-at-most-two pair count is$2r-3$ for a path of rank at least2, and$2r-2$ for D/E trees. Rank1 contributes0. The extra high-I_2 unit makes its total$2=2r-2$. D_5 alone has total$8+1=9=2r-1$; every other component contributes at most$2r-2$. These formulas follow directly by counting edges and pairs of neighbors; the checker verifies101 representatives and their exceptional marked pairs.

For a vertex of total rank $n \geq 6$, either its diagram is one component other than D_5, or it has at least two components, each with deficit at least1 from2r. Hence every vertex total is at most$2n-2$.

Double-counting planar angles and using the symmetric nonnegative h-vector gives a strict average lower bound

$$\frac{n(n-6)}{8} \quad (n \text{ even}), \qquad \frac{(n-1)(n-5)}{8} \quad (n \text{ odd}).$$

At n=21 the average weight is strictly greater than40, while the vertex upper bound is40. Explicitly,

$$f_2 - 50f_0 = \sum_{i=0}^{10}(10-i)(11-i)h_i > 0, \qquad 5 \cdot 50 - \binom{21}{2} = 40.$$

For $n=21+2u$ the non-strict lower-minus-upper difference is $u^2/2+5u$; for $n=22+2u$ it is $u^2/2+11u/2+2$. The strict face-count inequality therefore excludes every $n \geq 21$. This proves the stated bound using the published inputs and the independently reviewed geometric argument above.

The consequence for the previously established high-dimensional arithmetic field list is specific: the cubic-seven field disappears in dimensions at least22, leaving the two quadratic candidates. This does not eliminate either quadratic field or settle unconditional compact dimension29.

Verification scope

Generation and saved replay pass. The replay command is:

```
python scripts/fresh_compact_global_bound_20261015.py --verify
```

The checker imports SymPy for symbolic polynomial determinants. It imports no earlier producer arithmetic. It binds the frozen cubic-field package and compact h-vector package. The separate arithmetic/proof review now passes; the primary certificate's historical status string still says `PROOF_CANDIDATE` because its frozen source/data were preserved. No finite sample of stars is being substituted for the all-white-attachment reduction, and no external priority search completeness is claimed.

Source 7

Source path: `results/fresh_compact_global_bound_20261015_review.md`

SHA256: e05475299eff520a2c048aae27f884ad4beaf8ba3207915cade0414244191d4d

Independent review of the compact cubic-seven dimension-20 candidate

PASS within the stated compact arithmetic scope. The standalone checker and certificate independently reconstruct the finite graph reduction, polynomial rank identities, exceptional corner weights and dimension inequality in the primary proof. Generation and saved replay pass. This review finds no remaining mathematical gap in that proof, subject to its explicit published inputs.

The conclusion concerns compact hyperbolic Coxeter polytopes whose arithmetic ground field is $\mathbb{Q}(\cos(2\pi/7))$: their dimension is at most 20. It does not improve the unconditional compact bound, remove either quadratic arithmetic field, or claim priority. The previous dimension-28 fallback and its independent review remain frozen.

The new checker imports no producer code and uses no SymPy. A separate Fraction polynomial implementation expands both literal Gram determinants by sparse Laplace recursion, reduces powers using the cubic relation, and verifies every coefficient identity needed by the proof. It enumerates complete binary graphs instead of the producer's connected-shape/cross-edge loops. The earlier field classification, standard compact Coxeter face facts and the published arithmetic criterion are declared dependencies rather than claimed as new independent classification results.

Domain and geometric completeness

The global compactness assumption is essential and is used correctly. It excludes affine subdiagrams and supplies the actual-face correspondence for positive definite wall subsets. At a quadrilateral two-face, each adjacent black pair with the complete white support is spherical, whereas each opposite extension is indefinite. Its minimal indefinite diagram must contain both opposite black nodes, since white support plus either single black node occurs at a finite vertex. The inherited order bound therefore leaves exactly orders two and three.

The distinguished high bond in an order-three block does not require trusting the auxiliary seven-triangle enumeration. The inherited finite alphabet is $2, 3, 4, 6, 7, 14$. If a Lannér triangle had no 7 or 14 label, its determinant would lie in $\mathbb{Q}(\sqrt{2}, \sqrt{3})$; it could not generate a cubic ground field. This contradicts the same determinant-field proposition used in the fallback. Thus every such triangle has a $7/14$ bond. A bond of label at least six is isolated in any spherical diagram containing it.

The review enumerates all 35,904 binary graphs on the five possible minimal-block vertex configurations, including shared white nodes and every cross-bond choice. For each graph it checks connectedness of both minimal blocks, finite-corner forest constraints, isolation of each possible distinguished high edge, and absence of a marked path of length at most two. Exactly the producer's nine supports survive. They all have two distinct white nodes joined by an edge, with both high edges on opposite black pairs. In particular neither a dotted minimal pair nor a high black-white bond has been omitted.

The extension from these initial nodes to **all** white contacts is valid. Project to the orthogonal complement of the two positive definite opposite black blocks. A white node contacting either block has residual square below $-6/19$. A 14 block or a dense contact to both roots of a 7 block makes one square below -4. The cross product between white nodes attached to different blocks remains their old product, because the absence of short corners forbids cross contacts. Its square is less than one. The resulting two-negative-plane determinant is then greater than $5/19$, a contradiction. This excludes 14 and dense attachments at every white node, not only the initially chosen ones.

Let U, V collect all remaining white neighbors of the respective opposite black pairs. Every U-V edge is forced nonright: otherwise the two negative residual vectors would be orthogonal. The white diagram is a forest, so one part is a singleton.

A black endpoint cannot have two neighbors in the other part, since this would make a cycle in its finite corner diagram. Consequently the complete touched white set has sizes $1+1$ or $1+2$. This rules out hidden larger support patterns.

Every off-core white component attaches at most once to the connected white core; two attachments would produce a white cycle. Its connected edges have labels three or four, since label five is unavailable and a larger label cannot be attached to another edge in a finite diagram. Tree elimination therefore has rational diagonal corrections. A nonzero branch correction is strictly positive, and all reduced white diagonals remain positive and at most one.

Rank, rationality and the entire-component condition

The complete star has $|W|+4$ roots in a space of dimension at most $|W|+3$: the orthogonal complement of the white support is Lorentz three-space. After elimination of off-core white roots, the six- or seven-node matrix must therefore be singular. Only this upper rank bound is needed. A smaller local diagram with Lorentz signature but excessive rank is not a realizable two-face star.

The independently expanded six-node determinant is a nonzero rational quadratic in c^2, with leading coefficient the positive reduced white determinant. Since c^2 has degree three, this cannot vanish. This correctly removes the apparent path 7–!3–!3–!3–!7, which passes some local tests but fails the complete-star rank condition.

In the seven-node case the rationality claim also survives scrutiny. Squared bonds and reduced diagonals are rational. The one potentially irrational coefficient is $L = \delta\epsilon\beta\gamma \in \mathbb{Q}(\sqrt{2})$. The actual five-cycle has product a nonzero rational multiple of Lc. The arithmetic cyclic-product condition and $c \in K$ put L in K, whose intersection with the quadratic field is rational.

The separate Fraction determinant expansion gives the linear coefficient of the cubic remainder as $(\det H - 8L)/8$. Hence singularity forces $\det H = 8L$. Positive definiteness makes the determinant strictly increasing in each white diagonal, and therefore

$$\det H \le 1 - \delta^2 - \epsilon^2 \le \tfrac{1}{2} \le 8L.$$

Equality requires all three diagonals to be one and $\delta = \epsilon = \beta = \gamma = 1/2$. No nonzero eliminated branch can remain. The other coefficient then gives $\alpha = 1/2$. These claims are checked coefficientwise, not by numerical substitution at the cubic root.

The remaining four corners have two types D_5 and two types $A_4 + A_1$. Independent rational LDL tests certify all four. The marked D_5 pairs are precisely the two far-long-leaf/short-leaf pairs at distance three. All black-white contacts have already been exhausted by U, V, and all external white branches were removed by equality. Thus each weighted D_5 is an **entire connected component** of its full finite vertex diagram. This is the necessary justification for the half-weights; an arbitrary embedded D_5 would not suffice.

Global weight accounting and arithmetic scope

The triangle and quadrilateral lower weights are correct: a triangle contains a high rank-two corner receiving two units; a quadrilateral either has a short corner receiving one or the classified pair of D_5 half-weights. Larger polygons need only nonnegative weights.

The review reconstructs 122 finite diagrams. The all-rank proof follows directly from edge and neighbor-pair counts, not from that finite range. Every connected component has deficit at least two from $2r$, except D_5, whose deficit is one. At total rank at least six, a single component cannot be the exceptional D_5; two or more components have total deficit at least two. Hence the vertex weight is at most $2n - 2$.

The strict h-vector lower bound is applied with the correct parity. At dimension 21 it is strictly 40, matching the upper bound 40 and giving a contradiction. The independently reproduced gaps for all larger odd/even dimensions are $u^2/2 + 5u$ and $u^2/2 + 11u/2 + 2$. Strictness handles the zero constant in the odd case. The resulting bound is 20.

The arithmetic criterion and weighted-face/h-vector framework are the stated imports from Vinberg's original paper, §§1–3 and 7. This review is a reproducible mathematical and exact-arithmetic audit, not a formal proof-assistant verification or an exhaustive search for prior equivalent results.

```
python3 scripts/fresh_compact_global_bound_20261015_review.py --verify
```

Source 8

Source path: `results/fresh_raw357_coface_full_model_20261018.md`

SHA256: e15de1697e409773e3e567fc333e97a638bd0e20245904dc4624b73f861f5455

Sixteen coface transfers leave an exact lower bound above12

The sixteen existing-row transfers from the frozen raw357 coface theorem are now installed in a new copy of the complete current exported matrix. An exact feasible dual survives with lower bound

`4272824056360160 / 346500030107667 = 12.331381486554207…`

The inherited exact primal remains feasible, giving upper bound

`1233971157349 / 100000000000 = 12.33971157349.`

Therefore these sixteen transfers cannot lower the optimum of this finite model to12, even when all its face masses are free. No numerical optimization was needed to establish that obstruction. The lower certificate is a chosen feasible assignment, not an optimized assignment or an exact optimum.

The producer, certificate, and saved checker keep this model conclusion separate from geometric completeness. A feasible dual above12 is not a hyperbolic chamber, and no compact-arithmetic H13 exclusion or new dimension bound follows.

Exact changed matrix

The producer reconstructs the frozen all-marker model, including all prior ordinary charges and raw56/57/76/108/112 marker exports. It compares the complete prepared record with the frozen normalized-dual continuation. It then appends the sixteen coface columns, each of the form

`+1 on the raw357 source Q; −2b on its actual simplex S; vertex cost0.`

These are nonnegative transfer variables with signed face coefficients, not equalities or unrestricted variables. The simplex requirement remains20. The remaining31 simplex types absent from the inherited matrix are not installed, given artificial zero multipliers, or silently treated as present.

The new matrix has442226 variables,82806 face rows,6558 vertex-budget rows, 176400 active variables,96192 zero-budget signed variables and833711 face coefficients. Every old442210 variable,96176 zero-budget record, face coefficient and requirement, and vertex-budget coefficient is preserved literally. The52 old positive-budget signed variables and all14 multi-owner columns retain their complete costs. No unknown physical compatibility or historical owner assignment changes.

The exact matrix hash is 367da8a42e7b8f8cfd309f345486a7b7429f668657fcfdd60a57591422139948. Appending all sixteen valid columns does not assert that every one is new relative to every old constraint. The frozen coface certificate already establishes six strict violations by the old exact dual, so this is a genuinely strengthened model.

Cheap exact certificate before optimization

The old certificate has positive mass on nine raw357 source rows. Their required-payment-weighted mass is83284683/10000000000; each source has requirement1. Set these nine multipliers to zero and retain every other face multiplier and every vertex multiplier exactly.

This candidate satisfies all sixteen new inequalities: their positive source terms vanish and their simplex terms are non-positive. Deleting face mass could in principle invalidate old signed constraints, so the producer does not infer feasibility

solely from this observation. It reconstructs and checks every one of the442226 exact column inequalities against the full vertex-cost matrix. All pass. It also records that the nine changed face rows have no negative old coefficients, consistent with that result.

The objective numerator decreases by exactly83284683/10000000000, while the normalizer remains115500010035889/115500000000. The resulting rational ratio is the lower bound above. The old primal is verified on the same changed matrix with every new variable zero, giving the stated upper bound.

The model allows arbitrary nonnegative face and vertex multipliers; the chosen deletion is merely one feasible point. Weak duality makes its value a lower bound for the optimum over all such assignments. Thus changing other masses cannot yield a primal bound at most12 for this matrix. This does not assert that the optimizer would choose the deleted masses, nor that the improvement from the old optimum is known. The manual lower certificate is weaker than the previous lower certificate and is used because it remains feasible after the new constraints.

This gives a concrete next requirement: a further constraint must exclude this new feasible witness. A larger effect on the old, now infeasible raw357 masses alone is insufficient. Broader coface restrictions or other structural constraints should be tested against this surviving witness before another global solve.

Verification scope

Generation97599/94d717, separate full-matrix replay87103/8cef97, and the standalone saved checker b1ed83 all terminated exit0/PASS. No optimizer has been invoked.

Root completed a full producer/report mathematical accounting read. The construction reviewer completed a full producer/report/helper read and reread the inherited loads, cost-row and dual-verification interfaces. Both reviews passed: literal preservation, signed additions, exact mass deletion, normalization, complete feasibility obligations and the free-mass weak-duality conclusion are coherent. Neither reviewer executed the producer/helper, reconstructed the matrix, or repeated optimizer, geometry or historical census work. This report is final and frozen.

The producer rebuilds the exact inherited matrix and checks both complete certificates. It rechecks the bound saved charge arithmetic through the inherited builders; it does not repeat physical pair tests, native embedding censuses, clique enumeration or geometry proofs.

The independent checker imports no producer. It checks the complete sixteen saved transfers, the exact nine-row deletion, unchanged remaining multipliers, objective and normalizer arithmetic, inherited primal identity, model counts,14-owner preservation and interval arithmetic. It does not independently reconstruct the old matrix, establish old signed feasibility, or verify the primal against all old rows. Those obligations belong to the separate full producer replay. Historical geometry, catalogue completeness and the actual coface injection remain the frozen theorem inputs.

```
python3 scripts/fresh_raw357_coface_full_model_20261018.py --verify
python3 scripts/check_raw357_coface_full_model_saved_20261018.py
```

Source 9

Source path: `results/fresh_sqrt2_dotted_tail_20261015_review.md`

SHA256: 586960320dbecb75243b5ebf6b3503e299618495319810de96b5d2c65e8441ad

Independent review of the C5 pendant-tail lemma

PASS. The independent checker and certificate verify the primary classification, including its all-attachment argument within the stated topology. No gap was found. This is a local classification, not a complete quadrilateral theorem or a dimension bound.

The checker imports neither the producer nor a symbolic library. It uses a separate `Fraction` implementation of $\mathbb{Q}(\sqrt{2})$, permutation cycle-cover determinant expansion, congruence pivots, and exact residual geometry. The source/data bindings and every primary manifest entry are checked.

The all-attachment argument

Use the primary names: opposite pairs a, y and b, d, white chain $u - v - w$, the C5 $b - u - v - w - d - b$ with its sole 4-bond $b - d$, and contact $a - v$ of label 3. In the double family $y - v$ also has label 3. All other unlisted core pairings are right. Every extra connected white tree is attached by one edge to one core node; an orthogonal white component is harmless.

Both corners containing a have a D5 principal diagram. Its connected spherical extensions are simply laced D/E diagrams with only one branching vertex. Consequently a pendant contact at v creates a forbidden fourth arm, while a contact at u or w creates a second branching vertex in one of the two corners. At a, b, d, the only remaining possibility is one all-3 path. In particular, the first contact at b or d is **exactly** label 3, not merely at least 3. Together with the opposite 4-bond and the existing white contact it then gives the literal affine fork with arm labels 4,3,3. Compactness excludes it. The independent checker verifies its positive null vector and inertia.

The path at a gives D5,E6,E7,E8 for lengths 0,1,2,3; length 4 gives affine E8. This is checked by independently reconstructed matrices. Longer paths contain the forbidden prefix, and branching or a second attached path is already excluded by the finite-component classification. The double family gives the same restriction at y.

In the single family the y-component is separate from the C5 part in each finite corner. Its reduced unit diagonal q is positive and at most one. It is rational unless it is the isolated I2(8) component, which gives $q = (2 - \sqrt{2})/4$. Indeed every other permitted finite component has rational squared bonds, so exact leaf elimination remains rational; a label at least 6 cannot have a nonright neighboring edge in a finite diagram. Multiple trees at y are covered by treating their union with y as its one finite component. A nonzero attachment has strictly positive projection cost, so $q = 1$ forces no attachment.

These arguments cover arbitrarily large permitted pendant trees without enumerating their size. They do not cover an external component with two core attachments, extra core edges, shared core vertices, or changed core labels.

Independent rank and integrality checks

Permutation expansion of the seven-by-seven determinant reproduces both identities, with p, q the reduced unit diagonals and original norm-two contact $-r$:

$$\det G_{\text{single}} = 2(1 + \sqrt{2})\{r^2 - q(4p - 1 + \sqrt{2})\},$$

$$\det G_{\text{double}} = 2(1+\sqrt{2})\{r^2 + (1-\sqrt{2})r - 4pq + (1-\sqrt{2})(p+q)\}.$$

Independent inversion of each finite path gives $p = (\ell+2)/(2(\ell+1))$. Corank one is imposed on the **complete** affected star after all allowed extra white components have been eliminated. Orthogonal components contribute positive factors; a determinant calculation on an arbitrary smaller subdiagram would not suffice.

In the single family, the original two-cycle is

$$D = r^2 = q\left(\frac{\ell+3}{\ell+1} + \sqrt{2}\right).$$

If q is rational, integrality in $\mathbb{Z}[\sqrt{2}]$ makes its $\sqrt{2}$-coefficient q an integer. Thus $q = 1$, and the rational coefficient permits only $\ell = 0, 1$. All four exceptional I2(8) values are independently found nonintegral. The survivors have $D = 3+\sqrt{2}$ and $2+\sqrt{2}$, respectively.

For the double family, the original triangle $a-v-y$ has two contacts of magnitude one. Its cyclic product is $-r$, so integrality requires $r \in \mathbb{Z}[\sqrt{2}]$, a stronger requirement than integrality of r^2. Writing $r = A+B\sqrt{2}$ with integers A, B, the rank equation's $\sqrt{2}$-coefficient gives

$$2AB + B - A = p + q.$$

Among $p, q \in \{1, 3/4, 2/3, 5/8\}$, only $p = q = 1$ has integer sum. The checker evaluates all 16 pairs. At that pair the equation factors as $(r-1-\sqrt{2})(r+2)$, leaving the unique positive root $r = 1+\sqrt{2}$. This independently proves the integrality classification without using the producer's field-square-root routine. The review also checks every non-null contact recorded by that routine satisfies the equation and its recorded integrality status; it does not separately reproduce the intermediate nonintegral square-root membership decisions, which are unnecessary to this alternative proof.

The distinction between original and computational normalization matters. For single-family signature checks the producer replaces the original y by $\sqrt{D}\,y$, giving square $2D$ and contact $-D$. Since $D, D' > 0$, this is a valid congruence at both relevant embeddings. The arithmetic exclusions above concern the original norm-two two-cycle D, not cyclic products formed after that computational rescaling. The double-family cycle condition similarly uses the original unit-magnitude contacts.

Signatures, compact local faces, and remaining scope

For all three surviving matrices the independent checker reconstructs the physical and conjugate inertia, all four finite corners, and all **509** nonempty principal-subset tests excluding positive-semidefinite singular blocks. It additionally eliminates the white support, solves for all four residual vertex rays, and verifies negative norms, negative pairings with both nonincident sides, and negative pairings between all vertex rays. Thus the four finite corners are feasible and future-oriented in a common cone; their compact quadrilateral is not inferred from rank alone.

The indicated D5 marked long-leaf/short-leaf pairs and E6 marked near/far pairs agree with these literal corner graphs. Assigning half weights to the stated two pairs of an **entire** D5 or E6 component pays the three local survivors as described in the primary. This does not establish the corresponding inequality for other quadrilateral cores. External white components with multiple core attachments can produce Schur off-diagonal corrections and remain outside the present lemma.

Generation and saved replay pass:

```
python3 scripts/fresh_sqrt2_dotted_tail_20261015_review.py
python3 scripts/fresh_sqrt2_dotted_tail_20261015_review.py --verify
```

The complete result is exactly the primary's two fixed C5 attachment families with permitted pendant trees and orthogonal components. No global dimension improvement, complete ambient polytope, exhaustive quadrilateral classification, or novelty claim is certified.

Source 10

Source path: `results/fresh_sqrt2_lanner_catalogue_20261015.md`

SHA256: c278202d599bd2a2921f5d77fc3f6efa345b232489aabc67314660610856fe4b

Complete arithmetic Lannér input over $\mathbf{Q}(\sqrt{2})$

The standalone exact checker and certificate give the complete Lannér catalogue compatible with a compact arithmetic hyperbolic Coxeter chamber whose ground field is $K = \mathbb{Q}(\sqrt{2})$. There is one parametric order-two family, nine triangle types, and one type each of orders four and five. This is classification input for a separate quadrilateral analysis; it proves no new dimension bound.

All Gram matrices here have diagonal two. A finite bond m has negative entry $-2\cos(\pi/m)$. Dotted distances are not cut off or discretely sampled.

Published inputs and normalization

Vinberg's arithmetic criterion requires the norm-two Gram's cyclic products to be algebraic integers in the ground field, its entry field to be totally real, and its conjugate matrices away from the physical ground-field embedding to be positive semidefinite. The criterion is recalled in Nikulin, §2. A nonsingular principal block is therefore positive definite at such an embedding. The classification of Lannér diagrams, their maximum order five, and their role as minimal indefinite subdiagrams are given in Vinberg, §2.4, printed pages 74–75. These are published inputs, not conclusions of the finite enumeration below.

The square of a finite contact is $2 + 2\cos(2\pi/m) \in K$. Thus the real cyclotomic degree is at most two. The complete candidates are $2, 3, 4, 5, 6, 8, 10, 12$: the totient condition permits only primes 2, 3 and 5 with bounded exponents. Labels 5 and 10 generate $\mathbb{Q}(\sqrt{5})$, and 12 generates $\mathbb{Q}(\sqrt{3})$. The complete finite alphabet is consequently

$$\boxed{2, 3, 4, 6, 8}.$$

This is an all-label field argument, not an enumeration up to a chosen maximum label.

Order two: a complete parametric family

Write the Gram as

$$G = \begin{pmatrix} 2 & -\sqrt{t} \\ -\sqrt{t} & 2 \end{pmatrix}.$$

Its complete arithmetic conditions are

$$\boxed{t = a + b\sqrt{2} \in \mathbb{Z}[\sqrt{2}], \qquad t > 4, \qquad 0 < t' = a - b\sqrt{2} < 4.}$$

The first strict inequality is physical indefiniteness; integrality is the two-cycle condition. Total reality requires $t' > 0$, and conjugate positive definiteness requires $t' < 4$. The boundary values 0 and 4 cannot occur under a field embedding

of a nonzero t or $t-4$. Conversely these conditions give a totally real radical extension and the required two-cycle and conjugate-sign tests. The example $t = 3+\sqrt{2}$ is checked exactly. There is no finite catalogue of dotted magnitudes asserted.

Order three: exactly nine triangle types

All proper two-node subdiagrams are finite, so the finite alphabet applies to every bond. The triangle is physically hyperbolic exactly when $1/p + 1/q + 1/r < 1$. If a right angle occurs, the three-cycle vanishes. Otherwise its product must belong to K.

The magnitudes for labels 3 and 4 already lie in K. The other two are $\sqrt{3}$ and $\sqrt{2+\sqrt{2}}$, whose squareclasses are independent over K: 3 is not a square in K, and the norms of $2+\sqrt{2}$ and $3(2+\sqrt{2})$ are 2 and 18, both nonsquares in $\mathbb{Q}$. Hence each of labels 6 and 8 must occur an even number of times in a nonzero three-cycle. This exact parity condition tests field membership without symbolic numerical recognition.

For magnitudes a, b, c, the determinant is

$$\Delta = 8 - 2(a^2 + b^2 + c^2) - 2abc.$$

Each conjugate proper two-node block is positive definite. Therefore the remaining arithmetic sign test is exactly $\Delta' > 0$. All cyclic products that pass the field test are integral. The required radical fields are totally real because $3 > 0$ and $2 \pm \sqrt{2} > 0$.

Unordered labels	Δ	Δ'
2, 3, 8	$2-2\sqrt{2}$	$2+2\sqrt{2}$
2, 4, 8	$-2\sqrt{2}$	$2\sqrt{2}$
2, 6, 8	$-2-2\sqrt{2}$	$-2+2\sqrt{2}$
2, 8, 8	$-4\sqrt{2}$	$4\sqrt{2}$
3, 3, 4	$-2\sqrt{2}$	$2\sqrt{2}$
3, 8, 8	$-6-6\sqrt{2}$	$-6+6\sqrt{2}$
4, 4, 4	$-4-4\sqrt{2}$	$-4+4\sqrt{2}$
4, 6, 6	$-8-6\sqrt{2}$	$-8+6\sqrt{2}$
4, 8, 8	$-8-8\sqrt{2}$	$-8+8\sqrt{2}$

The certificate records all 35 unordered triples from the proved alphabet and the exact reason for every rejection.

Orders four and five: one cycle each

An edge of label at least 6 in a connected diagram has an adjacent nonright edge. Their three-node Gram has determinant no greater than the relaxed unit-square value $1 - \cos^2(\pi/6) - \cos^2(\pi/3) = 0$. Any additional bond only decreases the determinant. For Lannér order at least four this is a proper subdiagram, contradicting positive definiteness. Parallel and dotted bonds likewise contradict proper two-node positivity. With the field alphabet, only labels 2, 3 and 4 remain.

A tree with these labels cannot be arithmetic over K: conjugation changes the signs precisely on the 4-bonds, and vertex sign changes on a tree undo all those changes. Its conjugate Gram is therefore congruent to its indefinite physical Gram.

If a Lannér graph contains a cycle, that cycle must use all vertices. Indeed the all-3 cycle already has a nonpositive all-ones quadratic form, and larger bonds or additional chords decrease it. A proper cycle would violate proper-subdiagram positivity. There can be no chord, because it produces a shorter cycle. The graph is consequently a chordless cycle.

An even number of 4-bonds again makes the conjugate Gram sign-switch-congruent to the physical Gram. An odd number at least three in a cycle of length four or five forces two adjacent 4-bonds; their proper three-node path is affine. Exactly one 4-bond remains. Both cycles pass all proper-subdiagram and conjugate tests:

Order	Diagram	Norm-two determinant	Conjugate determinant
4	C4 with labels 4,3,3,3	$-1-2\sqrt{2}$	$-1+2\sqrt{2}$
5	C5 with labels 4,3,3,3,3	$-2-2\sqrt{2}$	$-2+2\sqrt{2}$

As a separate check of this analytic classification, the script tests every labelled graph on four and five nodes after the proved 2/3/4 reduction. It checks every maximal proper principal matrix exactly, then the physical determinant and the conjugate Gram. The 729 and 59,049 graphs give respectively 12 and 60 labelled arithmetic survivors, each one isomorphism class. Six additional labelled order-four Lannér graphs fail the conjugate test. No numerical eigenvalue acceptance or project arithmetic import is used.

What a complete no-short-corner argument must still handle

For a compact quadrilateral with full white support W and cyclic black sides a, b, c, d, each opposite block $W + a + c$ and $W + b + d$ is indefinite. It contains a minimal Lannér diagram that includes both of its black nodes. Thus the list above is the complete input for those two chosen pieces. It does not classify their union or the complete quadrilateral star.

The frozen disjoint-cycle bridge theorem handles two disjoint single-4 cycles with one direct white bridge and certain pendant white trees, provided at least one cycle has order four or five. Beyond that theorem, a global weighting proof still needs the following reductions:

- One or both opposite pieces can be a dotted pair, including the entire unbounded algebraic family above. All nine triangle types must also be retained, not just the 3,3,4 cycle.
- The selected pieces can share white nodes. Their union can have further cross contacts; one cannot assume disjointness or a unique bridge before a finite-corner argument proves it.
- White vertices outside the chosen union can attach at black nodes or at several parts of the union. Eliminating such a component can change off-diagonal entries as well as diagonals. The pendant-white theorem's rational diagonal correction is not automatically valid for those attachments.
- The complete star, rather than an arbitrarily selected subdiagram, must satisfy the ambient rank defect. Full-rank local Lorentz subdiagrams can sit inside a larger singular star. A determinant calculation on a partial union alone is not a contradiction.
- Every proposed weighted exceptional corner must be the entire spherical component whenever its weight bound depends on that property. The simultaneous four actual-corner inequalities must be checked, not only the selected Lannér blocks.

For example, the already frozen singular path with labels $8, 3, 4, 3, 8$ and the singular bridge of two 3,3,4 cycles show why the cubic-seven quadrilateral argument does not transfer unchanged. Both have relevant F4 corners. They motivate testing an entire-F4 correction, but the present catalogue does not establish that correction's quadrilateral inequality.

The separate quadrilateral support enumeration is owned by the compact-search lane. This package does not duplicate it and makes no exhaustion claim about external attachments.

Reproduction

```
python3 scripts/fresh_sqrt2_lanner_catalogue_20261015.py
python3 scripts/fresh_sqrt2_lanner_catalogue_20261015.py --verify
```

Generation and saved replay pass. Primary literature was checked on 2026-10-15 for the inputs cited above. No novelty or current-best claim is made.

Source 11

Source path: `results/fresh_sqrt2_quad_exhaustion_20261015.md`

SHA256: 3c5d541e44dd8afa9b0f97e335f102d26292649ad11932c72a68ae523e00ce73

A complete quadrilateral reduction over $\mathbf{Q}(\sqrt{2})$

Internally independently reviewed proof candidate: a compact arithmetic hyperbolic Coxeter polytope with ground field $\mathbb{Q}(\sqrt{2})$ has dimension at most **20**. The finite calculations below are complete within their stated roles. The argument for arbitrarily many external white nodes is given separately; a cutoff on those nodes is not used. No unconditional compact bound or bound for other arithmetic fields is asserted. Internal verification does not establish literature priority or replace external mathematical review.

The computation consists of a minimal-pair union enumeration with data, an exact arithmetic/signature filter with data, a complete one-extra-white contact calculation with data, and a closure/weight checker with data. The independent global review reconstructs the union enumeration, marked orbit coverage, exact core/profile arithmetic and closure identities without producer imports, and separately audits the all-size proof.

Published inputs and the finite reduction

We use the usual compact Coxeter facts: a vertex diagram is spherical; every elliptic subset corresponds to an intersecting collection of facets; affine principal subdiagrams are absent; and a minimal indefinite subdiagram is Lannér. Arithmetic cyclic products lie in the ground field and are algebraic integers; every other ground-field embedding makes the Gram matrix positive semidefinite. These are inputs, not new conclusions of the search. Vinberg, §§1–3 and 7.

The complete $\mathrm{Q}(\sqrt{2})$ Lannér catalogue proves that the finite alphabet is $2, 3, 4, 6, 8$. Its minimal indefinite pieces are a dotted pair, nine triangle types, or a chordless C4/C5 with exactly one 4-bond and all other bonds 3. The nine unordered triangle types are

$$(2,3,8),(2,4,8),(2,6,8),(2,8,8),(3,3,4),(3,8,8),(4,4,4),(4,6,6),(4,8,8).$$

The dotted distance remains an arbitrary permitted real arithmetic parameter. It is not replaced by a finite list.

Let W be the **full** white support of a quadrilateral two-face, and label its black sides cyclically $0, 1, 2, 3$. Every $W + \{a, b\}$, with $a \in \{0, 2\}$, $b \in \{1, 3\}$, is spherical. The opposite extensions $W + \{0, 2\}$ and $W + \{1, 3\}$ are indefinite, and contain minimal Lannér subdiagrams L, M containing their respective two black nodes.

If any marked adjacent black pair has graph distance at most two in its full finite corner, the basic weight below already pays for this quadrilateral. Assume henceforth that no such short marked path exists. Enumerate $L \cup M$, not all of W. Each piece uses at most three white nodes, so this union has at most six whites. Both shared white nodes and all additional contacts on the union must be considered.

The enumerator includes:

- all 51 marked piece representatives, fixing the ordered two black labels and quotienting only white names;
- all 6,010 shared-white identifications, with injective identifications and canonical names for new whites;
- every unspecified union contact in the complete finite alphabet;
- all four exact finite-corner tests and absence of a marked path of length one or two.

All pairs except the two opposite black pairs lie in at least one finite corner and hence really do have finite labels. The opposite pair in each piece is already specified by that piece, possibly dotted. There are no unenumerated continuous cross contacts. Interchanging the two chosen pieces is just a relabelling of an abstract quadrilateral, not an assumed symmetry of an ambient polytope.

The traversal terminates without a cutoff after 12,746 nodes. It gives **115 labelled union supports**, or **52 marked isomorphism classes**. Every deduplication has a saved full-pair bijection preserving the black nodes, opposite relation and labels. Finite tests use exact rational forest Schur elimination. Cycles are nonpositive already at their least allowed bonds; an edge of label 6 or 8 in a finite diagram must be an isolated rank-two component.

Disjoint orthogonal Lannér pieces are rejected because their spans supply two negative directions. Otherwise all extra union contacts are retained; no bridge-only assumption is made.

Exact principal-subdiagram obstructions

The signature filter works in

$$\mathbb{Q}(\sqrt{3}, \sqrt{2+\sqrt{2}}),$$

using exact algebraic-field matrices and rational isolation of its real primitive element. For each finite-label core it eliminates a positive-definite corner and tests the remaining two-by-two Schur matrix. Both physical diagonals are negative. A positive determinant therefore proves two negative directions, independently of every omitted wall.

The other embedding sends $\sqrt{2}$ to $-\sqrt{2}$, and $\sqrt{2+\sqrt{2}}$ to $\sqrt{2-\sqrt{2}}$. Its Schur diagonals and determinant must be nonnegative. Fundamental-cycle parity also checks that cyclic products belong to $\mathbb{Q}(\sqrt{2})$; an odd occurrence of the independent label-6 or label-8 radicals violates this condition.

Of the 52 classes, **40 have two negative directions and one fails the cyclic-field condition**. The remaining classes are two dotted cores and nine finite-label cores. The latter have IDs

$$5, 9, 14, 17, 32, 37, 41, 47, 50.$$

These IDs refer to the explicit saved class list. Seven are singular; classes 14 and 37 are nonsingular. A principal core passing these tests is not yet a complete two-face star.

Complete one-white contact domains

Any additional white node must have one of the finite contact patterns checked against the union core. For the nine finite-label cores, exact finite-corner and short-path tests leave **44** nonempty patterns. The field, physical-signature and conjugate-positive-semidefinite tests leave only **four**:

Core	Permitted nonempty contact to one new white node
5, 9, 17, 32, 41, 47, 50	none
14	label 3 to black node 2 only
37	label 3 to black node 0 only; to black node 2 only; or to both

All other contacts in each accepted row are right. For singular conjugate cores, every new row must annihilate the exact nullspace; this rejects 24 rows. The other rejections are saved with their exact field or Schur witnesses. The calculation imposes no bound on the number of further white nodes.

The seven singular cores are therefore **sealed**: every other white node is orthogonal to every node in such a core. Their affected finite corner components are consequently entire components, regardless of how many orthogonal white components exist elsewhere.

For the two dotted cores, the finite-corner calculation is independent of the dotted value. It leaves 12 and 6 patterns respectively. Apart from contacts to the two dotted endpoints, the only possibilities are a new 3-neighbor of black node 1, a new

3-neighbor of black node 3, or a common 3-neighbor of both. A single such neighbor produces a global affine fork with arm labels $4, 3, 3$, forbidden by compactness. A common neighbor gives a $(3, 3, 4)$ Lannér triangle orthogonal to the dotted pair, hence two negative directions. Thus every external white node can contact the dotted core **only at its dotted endpoints**. This includes the common-neighbor case; it is not silently discarded as a pendant attachment.

Arbitrarily large white supports: the finite-label cores

For core 14, a complete star cannot have no external contact: its nonsingular connected Gram would remain nonsingular after adjoining only orthogonal positive white blocks, whereas a complete quadrilateral star must have corank one. Its sole possible contact is a 3-neighbor at node 2. Adding that node gives a literal marked copy of sealed core 17. No further white node can contact this larger core. This closes every extension of class 14, including paths through whites initially having no contact with the old core.

For core 37, write $a = 0, y = 2$. Its opposite contact is label 4, and both endpoints touch the middle of the fixed C5 by label 3. Every external contact is at a and/or y, by the table. The white graph is a forest. An external connected component cannot have two contacts to the same endpoint: that would create a cycle in a finite corner. Two distinct components at the same endpoint would make a second branching vertex alongside the existing D5 branch, also impossible in a finite diagram.

In a corner containing a, an attached component extends the short arm of D5. A finite connected diagram containing this branch is simply laced of type D or E, has only one branching vertex, and permits only a simple path of length $0, 1, 2, 3$ at that arm. These yield D5, E6, E7, E8. Length four contains affine E8. The same statement holds at y. This classification is the all-size attachment argument; the four finite cases and first affine obstruction are exact checks, not a size cutoff.

If an external component connects a to y, its white part is therefore a path with one, two or three nodes. Together with the opposite 4-bond it gives a C3, C4 or C5. The existing C5 and all extra old contacts remain present. The resulting full union is respectively a saved literal copy of sealed core **41, 47 or 50**. The checker records the complete marked pair bijections. Hence these extensions also stop without further attachments.

If no component connects a and y, only separate pendant paths remain. Eliminating them changes the two unit diagonals to

$$p, q \in \left\{1, \frac{3}{4}, \frac{2}{3}, \frac{5}{8}\right\}.$$

For the original norm-two opposite contact $r = \sqrt{2}$, direct full-star rank gives

$$(p + q - 4pq) + (1 - p - q)\sqrt{2} = 0.$$

Both coefficients must vanish, forcing $p + q = 1$, while the allowed tails give $p + q \geq 5/4$. Thus a nonsingular class-37 core cannot be completed only by such pendants. This covers all its external components, including branches and paths through vertices not initially adjacent to the core.

Arbitrarily large white supports: the dotted cores

Both remaining cores have a C5 (b, u, v, w, d, b) with its only 4-bond on b, d, and a dotted pair a, y. Node a touches v by a 3-bond; node y is either right to v or also touches it by 3. The complete one-white calculation proved that no external contact is possible except at a, y.

If no external white component connects both endpoints, every component is pendant at one endpoint or orthogonal. The independently reviewed pendant classification then applies without an attachment cutoff. It leaves only the untailed single-contact dotted star, its one-tail label-8 version, and the untailed double-contact dotted star. Their affected complete components are D5 or E6. The label-8 version is already sealed core 17.

It remains to exclude a white component connecting both dotted endpoints. Eliminate **all** external white nodes. Their matrix is positive definite with nonpositive off-diagonal entries, so its inverse is entrywise nonnegative. In unit normalization the resulting endpoint diagonals satisfy $0 < p, q \leq 3/4$: each endpoint had at least one nonright contact of cosine at least $1/2$, and the corresponding diagonal inverse is at least one. The off-diagonal correction is strictly negative because the two endpoint contact supports meet the same connected white component. Thus the effective norm-two opposite magnitude satisfies

$$r_{\text{eff}} > r > 2.$$

The endpoint bonds of the connecting component need not all be 3; labels 4, 6 or 8 at an endpoint only strengthen these inequalities.

The two exact rank identities from the pendant package remain valid for these reduced diagonals and increased effective contact, because elimination creates no contacts elsewhere in the core. In the single-contact case,

$$r_{\mathrm{eff}}^2 = q(4p - 1 + \sqrt{2}) \le \frac{3}{4}(2 + \sqrt{2}) < 4,$$

contradicting $r_{\mathrm{eff}} > 2$.

For the double-contact case put $t = (1 - \sqrt{2})/2 < 0$. The positive root is

$$r_{\mathrm{eff}} = -t + \sqrt{(2p - t)(2q - t)}.$$

It increases with both positive variables. Therefore

$$r_{\mathrm{eff}} \le \frac{1}{2} + \sqrt{2} < 2,$$

again a contradiction. The other root is negative. This excludes all multiply attached exterior components and justifies reduction to the reviewed pendant theorem.

We have now exhausted every compact quadrilateral with no distance-two marked corner. No assumption that the white support was already contained in the initial minimal-pair union remains.

Face weights and the dimension consequence

Assign weight 1 to every marked pair at graph distance at most two in its finite vertex diagram. Add the following nonnegative bonuses, **only when the indicated connected component is entire**:

Entire finite component	Additional marked pairs and weights	Total bonus in that component
F4	the two ends of the path, weight 1	1
D5, arms 1,1,2	long-arm tip with either short-arm tip, weight 1/2 each	1
E6, arms 1,2,2	near node on one length-two arm with far node on the other, weight 1/2 each	1
E7, arms 1,2,3	near node on the length-three arm with the length-two-arm tip, weight 1/4	1/4
E8, arms 1,2,4	near node on the length-four arm with the length-two-arm tip, weight 1/4	1/4

The exact corner check gives the following payments for the sealed cores: class 5 has one F4 corner; class 9 has two; class 32 has four; class 17 has two E6 corners; class 41 has four; classes 47 and 50 have four E7/E8 corners respectively. The dotted stars have two or four D5 corners. Thus every exceptional quadrilateral receives at least one unit. Other quadrilaterals already received one from a short marked pair.

Every triangular face receives at least two units from the basic distance-two rule. A minimal Lannér subset of its full star contains all three black nodes. If it has order three, its connected graph has at least two nonright edges. For C4/C5 the three cyclic gaps between the black nodes are positive integers summing to four or five; at least two gaps have length at most two. Their paths have only white internal nodes, so each occurs in the corresponding finite corner. Larger faces receive nonnegative weight, which is at least $5 - k$ for $k \ge 5$. Therefore every two-face with k sides has total weight at least $5 - k$.

For an irreducible finite component of rank r, the basic count is $2r - 3$ on a path and $2r - 2$ on D/E trees. After bonuses, D5 and E6 have deficit $2r - w = 1$; E7/E8 have deficit $7/4$; all other components have deficit at least two (A1 also has deficit

two). For total rank $n \geq 9$, a single component has deficit at least two, because all exceptional components have rank at most eight; two or more components have total deficit at least two. Consequently every vertex receives at most

$$2n - 2.$$

The standard h-vector/face-incidence inequality used in the reviewed compact-bound package makes the average vertex weight strictly greater than

$$\begin{cases} n(n-6)/8, & n \text{ even}, \\ (n-1)(n-5)/8, & n \text{ odd}. \end{cases}$$

For odd $n \geq 21$, the lower expression minus $2n - 2$ is $(n-1)(n-21)/8 \geq 0$. For even $n \geq 22$, it is $(n(n-22)+16)/8 > 0$. Strictness also excludes equality at $n = 21$. This gives the stated dimension-at-most-20 consequence. The separate weight certificate and its independent review also check the complete component charges and scalar inequality.

Verification scope

All four primary generation commands and their `--verify` replays pass. Independent generation and saved replay also pass. The finite signature and extension tests use exact number-field arithmetic, not numerical eigensolver tolerances. The source manifests bind the complete Lannér catalogue and the independently reviewed dotted-tail theorem. The unbounded white-component argument and the h-vector implication are mathematical proofs, not extrapolations from the finite graph tables. The immutable closure JSON retains its historical pre-review status string; the linked, separately hash-bound review records the completed independent validation.

No earlier frozen source or certificate is modified. Independent hostile review of the complete support enumeration, attachment closure and final scope reports no gap. This package is now frozen.

Source 12

Source path: `results/fresh_sqrt2_weight_bound_20261015_review.md`

SHA256: 47f434698cca5169ca427b77a63cf10fcdb1be37f49e32f1b2c16ae32c8b9c5e

Independent review of the Q(sqrt2) weight accounting

The proposed weights and their **conditional dimension-20 implication pass** an independent reconstruction. The separate global proof must still show that every quadrilateral, including all external white components, reduces to the listed entire-component configurations. This review does not establish that reduction.

The standalone checker imports no producer, project arithmetic, NetworkX, or graph-isomorphism routine. It uses breadth-first distances, an intrinsic arm-length classifier, and exact Fraction arithmetic in Q(sqrt2). The certificate records all literal corner diagrams, finite Schur pivots, weights, component totals, partition maxima, and dimension inequalities.

Literal corner reconstruction

All seven selected singular finite-label cores are reconstructed from the raw labelled union lists. Their four actual finite-corner graphs are tested by exact leaf elimination: after normalizing diagonal entries to 2, only the squares of the edge entries are needed, so the arithmetic remains in Q(sqrt2), including label 8. This is an independent positive-definiteness check, not only a count of graph distances.

For the three pendant survivors, the checker starts with the saved **scaled physical Gram matrices**, computes every corner's normalized Cartan cycles, and recovers the finite labels. It does not reuse the primary weight checker's handwritten tail graphs. All twelve tail corners agree, with quadrilateral totals **1, 1, 2**. Each of the seven finite-label cores likewise has total at least 1, with all four individual weights matching the primary certificate.

The corrections are intrinsic to complete connected components. Equal-length arms receive symmetric assignments; the E7/E8 distinguished arms have different lengths; and the F4 endpoints are intrinsic. Thus the assignments are invariant under all diagram automorphisms. The code also verifies a relabelling guard. No correction is assigned merely because a proper subdiagram resembles one of the exceptional types. This entire-component requirement is essential to the separate classification dependency.

All ranks and triangular faces

The base distance-two pair count is $2r - 3$ on a path of rank $r \geq 2$, and $2r - 2$ on a D/E tree. These formulas follow directly by counting edges and unordered neighbor pairs. A rank-one component contributes zero. After the corrections, the only component deficits from $2r$ below 2 are:

Component	Total weight	Deficit
D5	9	1
E6	11	1
E7	49/4	7/4
E8	57/4	7/4

F4 has weight 6 and deficit 2. Every other finite component has deficit at least 2. Consequently, at total rank $n \geq 9$, a single component has deficit at least 2; two or more components also have combined deficit at least 2, because each component

has deficit at least 1. This proves the all-rank vertex bound $2n - 2$. The additional exact partition calculation through rank 100 attains that bound for every $n \geq 9$; it is a check of the proof, not a finite cutoff standing in for it.

For a compact triangular face, a minimal indefinite diagram in its full star must contain all three marked black nodes: omitting any one would put it in a finite corner. The separately supplied Q(sqrt2) Lannér catalogue restricts this diagram to a connected triangle or a cycle of length four or five. The connected triangle supplies at least two nonright edges. In a four- or five-cycle, the three positive cyclic gaps between black nodes sum to at most five, so at least two gaps have length at most two. Each of these paths avoids the third black node and is present in the relevant finite corner. These give two base weights, regardless of any additional nodes. Hence every triangle has weight at least 2. All cycle markings are independently checked. The catalogue's global completeness remains a named input, rather than an inference from those markings.

Strict dimension inequality

For a simple compact n-polytope, let $h_0, \ldots, h_n$ be the symmetric nonnegative h-vector of its simplicial dual. Then

$$f_0 = \sum h_i, \qquad f_2 = \sum_i \binom{n-i}{2} h_i.$$

After symmetry, the coefficient of h_i in f_2 is $\{\binom{i}{2} + \binom{n-i}{2}\}/2$. Its minimum is $n(n-2)/8$ for even n and $(n-1)^2/8$ for odd n. Since $h_0 = h_n = 1$ and the two endpoint coefficients are strictly larger than the minimum, f_2/f_0 is **strictly** larger than this minimum.

Conditional on quadrilateral weight at least 1, every k-gonal two-face has weight at least $5 - k$: triangles and quadrilaterals give the nontrivial cases, while nonnegative weights suffice for $k \geq 5$. Double-counting angles gives

$$\frac{\text{total vertex weight}}{f_0} \geq 5\frac{f_2}{f_0} - \binom{n}{2} > \begin{cases} n(n-6)/8, & n \text{ even}, \\ (n-1)(n-5)/8, & n \text{ odd}. \end{cases}$$

For $n = 21 + 2u$ and $n = 22 + 2u$, subtraction of $2n - 2$ gives respectively $u^2/2 + 5u$ and $u^2/2 + 11u/2 + 2$. Both are nonnegative for every integer $u \geq 0$. The first is zero at n=21, where the strict h-vector inequality still contradicts the vertex upper bound 40. Thus the conditional conclusion is n at most 20, with no parity or equality gap.

The finite spherical diagram classification, h-vector facts, and the separately supplied Lannér catalogue are the mathematical inputs. Exhaustion of all quadrilateral stars and arbitrary external attachments is outside this review. The arithmetic and accounting do not by themselves promote a bounded local census into an unconditional theorem.

Generation and saved replay pass against the final primary dependency hashes. Source manifests, syntax, and local links pass. No mathematical discrepancy was found.

Source 13

Source path: `results/fresh_sqrt5_dimension23_coverage_20261015.md`

SHA256: 365d097f3ef03383ff1127f2ee0b03c9b5396fb2e754a8b45997366a6dcc3c6e

Compact arithmetic Q($\sqrt{5}$): whole-component hub weights

The source and complete certificate pass generation and saved replay. They bind the completed preceding coverage, the final independently reviewed finite queue and the independent whole-component weight review.

The certified statement is precise: a compact arithmetic hyperbolic Coxeter polytope with ground field exactly Q($\sqrt{5}$) has dimension at most 23. The preceding binder excludes dimensions 25–29. This separate composition excludes dimension 24 using the whole-I2(5) rule and its independent exact review. No previous certificate is changed.

Exact whole-vertex counts

Let a be the number of entire A1 components and b the number of entire I2(5) components in a spherical vertex diagram. Connected-pair bonuses and the normalized A1 cross bonuses depend on a and on whether b is zero. A disconnected A1-to-component angle receives u/a, as before.

The additional cross family joins a root of an entire I2(5) to a root in another entire component, whose type is neither A1 nor I2(5). Such an angle receives v_a/(2b). For each destination root there are exactly 2b source roots, so their sum is v_a. A1-to-I2(5) pairs belong only to the A1 family; I2(5)-to-I2(5) pairs receive no cross bonus. Thus the two families do not overlap, and there is no double counting. The root-orbit multiplicities and both denominator cancellations are verified independently.

For fixed exact (a,b), the remaining forest contains neither of these two component types. The independent type-first dynamic program sums its additive component costs, then adds the a A1 and b I2(5) costs. All 157 feasible contexts in rank 24 are included, with all 77 allowed finite component types. The exact maximum is

`267527/5000 = 53.5054 < 54`,

with margin `2473/5000`. This is a feasible upper bound, not an optimality claim. Floating-point optimization only proposed the rule; the displayed bound and all payments use exact rational arithmetic after upward grid rounding.

Face payments and arbitrary spectators

The short-path and entire-H4 baseline is unchanged. All bonuses are nonnegative. The independently reviewed 76 triangle markings still receive at least two, with the same spherical-extension stability. Any persistent short/H4 quadrilateral payment is also retained.

For a k-node zero-touch core, the external rank in dimension 24 is 26−k. The binder uses one common pair (e,f) of external A1 and I2(5) counts at all four corners. It checks every nonnegative pair with `e+2f≤26−k` and remaining rank zero or at least two. The remaining forest has neither A1 nor I2(5); A2, B2 and A-type components show that every rank at least two is possible for this conservative enumeration. A remainder of one is impossible. Each corner's own counts are added before evaluating its presence flag and denominators.

The complete-weight independent review directly checks 2,516 prior-catalogue/full-finite-queue rows in 65,300 spectator cases. It includes all 1,356 rank-feasible finite singular stopping states from the complete 2,070-obligation list. Expanded and previously paid states are not omitted. Its minimum face slack is 1/170000.

The current binder additionally replays every older coverage branch under this new rule: all original sealed cores, all 147 singular first attachments, their representative/subcore maps, the 51-state singular frontier, all seventeen continuation-closure states, and every rank-feasible dotted-queue singular state. Every finite-queue singular row is also reevaluated directly. An over-rank row is skipped only for the dimension-24 payment; it remains part of the complete arithmetic queue and the preceding higher-dimensional coverage. No old long-pair payment is assumed to transfer.

The arithmetic and geometric completeness argument is unchanged: conditional sealing makes marked components entire unless a touching root is present; every such root is followed through the reviewed exact domains or creates a persistent baseline payment. Orthogonal whites are precisely the spectators just enumerated. Nonsingular no-touch cores are rank-excluded. The complete finite and dotted queues use the justified 31-node ceiling for dimensions at most 29, which also covers dimension 24. All shared-white minimal unions, exact dotted parameters, literal root/child maps and singular stopping alternatives remain source-bound dependencies of the preceding proof.

Strict dimension-24 contradiction

With every triangle paid two, every quadrilateral one and all weights nonnegative, double counting gives average vertex weight at least $5\ f_2/f_0 - \binom{24}{2}$. The symmetric nonnegative h-vector of a simple compact polytope gives $f_2/f_0>66$: the minimum paired coefficient is 66, while $h_0=h_{24}=1$ contributes a strictly larger extreme coefficient. Hence the average vertex weight is strictly greater than $5\cdot 66-276=54$.

This contradicts the exact upper bound 267527/5000. Together with the completed preceding coverage and the known compact bound 29, it yields the stated field-specific conclusion. Other fields and noncompact or nonarithmetic classes require separate arguments; this certificate makes no claim about them or about priority.

```
python3 scripts/fresh_sqrt5_dimension23_coverage_20261015.py
python3 scripts/fresh_sqrt5_dimension23_coverage_20261015.py --verify
```

The final certificate records 2661 zero-touch rows, of which 1897 fit dimension 24, and checks 60735 common spectator-count cases. It verifies 471 literal maps and 73 recursive manifests containing 722 source-hash references. Generation, saved replay, syntax and local-link checks pass.

Source 14

Source path: `results/fresh_sqrt5_dimension24_coverage_20261015.md`

SHA256: 2c55cb61fa72bdb8534f665622d2fd8fe6e86c52e5eaa28335132c1115b802c0

Complete-face composition for compact arithmetic $\mathbb{Q}(\sqrt{5})$

The binder and complete certificate pass generation and saved replay. They bind the completed dimension-27 coverage certificate, the final independently reviewed arithmetic queues and the three separate exact weight reviews.

The certified conclusion is: every compact arithmetic hyperbolic Coxeter polytope whose ground field is exactly $\mathbb{Q}(\sqrt{5})$ has dimension at most 24. The preceding coverage binder excludes dimensions 28 and 29; this composition adds dimensions 25, 26 and 27. The previously established compact upper bound 29 covers all higher dimensions. Other fields, noncompact polytopes and arbitrary nonarithmetic polytopes are outside this statement.

Rules on whole vertex diagrams

Use the three complete-queue weight rules, with their separate exact implementation review. Their common baseline puts one on pairs at graph distance one or two and the reviewed extra units on the 5-edge and endpoint pair of an entire H4. All additional weights are nonnegative.

Connected-pair bonuses depend on the entire connected spherical component, its marked pair and the exact number a of entire A1 components in the vertex. A disconnected pair joining an A1 to a root of another entire component receives u_a/a. No A1–A1 pair is charged. For each root of the other component there are exactly a such pairs, so its total contribution is u_a. This justifies additive component totals for every fixed a; the separate review verifies the multiplicities and this normalization literally.

The reviewer recognizes labelled finite diagrams intrinsically, including the A2/I2(3) and B2/I2(4) aliases. Its exact type-first rank dynamic program is independent of the optimizer and of the producer's rank-first recurrence. It checks all finite component types through the target dimension and all realizable exact A1 counts. The sole impossible count is $n-1$, since the remaining rank one cannot be a non-A1 component. No numerical optimization status or optimality assertion is used.

Dimension	Exact maximum vertex weight	Strict lower threshold
25	59	60
26	31,017/500	65
27	67,009/1,000	143/2

The current binder independently recalculates these maxima from the literal rational rules and the reviewed finite-component implementation. It also rechecks all 76 minimal triangle markings using the weaker H4-only baseline. Every triangle gets at least two, and this persists under allowed spherical extensions. Nonnegative bonuses preserve all existing short-path and entire-H4 quadrilateral payments.

Complete coverage with the current weights

The geometric and arithmetic reductions are those of the preceding complete coverage proof. An unpaid compact quadrilateral yields two minimal arithmetic Lannér pieces containing its opposite black pairs. All shared-white and finite cross contacts occur in the reviewed 5,440-support/1,180-class reduction. After field, signature and persistent-payment filters, the

remaining cases are the 112 sealed singular finite cores, 44 sealed nonsingular rank exclusions, 130 finite cores with 458 possible first attachments, and 24 dotted parameter families.

The new weights are not assumed to preserve any old long-pair payment. The binder reconstructs all 458 literal first attachments, every sealed-subcore and representative map, all 147 singular first attachments, all 51 marked singular frontier states and all seventeen closure states. The seven touching continuations are explicitly mapped to their closure representatives. Each row is paid again under each current dimension-specific rule whenever its corner rank fits that dimension. The same direct check is made for all 264 singular dotted-queue states and all 2,070 singular finite-queue states.

The finite queue has 5,172 complete marked states from all 311 nonsingular literal starts, with no pending state. Every singular stopping choice is included, even if it admits further attachments or was paid by an earlier rule. Nonsingular no-touch stars are rank-excluded; every admissible next touching root is followed unless a persistent baseline payment or a necessary arithmetic obstruction ends that branch. The separate full queue review is a required source dependency. The dotted queue's 438 states cover all 32 exact parameter choices in the 24 families and likewise retain every singular stopping choice.

A full quadrilateral star in dimension n has n+2 roots. Thus the queue's 31-node ceiling covers every target $n \le 29$ and is a dimension bound, not a search cutoff. For a displayed k-node singular core, its corner rank is $k-2$. It contributes no case at dimensions below that rank. Otherwise the external rank is $n-k+2$: every possible exact number e of external A1 components is checked, excluding only a leftover non-A1 rank of one. No assumption of sealing is made for an intermediate singular stopping case: it is paid only when the remaining white roots are orthogonal spectators, while all touching alternatives continue through the queue.

Conditional sealing is used exactly as in the reviewed branches. If every nonzero next-white contact is excluded or creates a persistent payment, all uncharged outside roots are orthogonal to the core. Its marked components are then entire. Orthogonal spectators may be arbitrary spherical forests, and their exact A1 contribution is included. A sealed nonsingular star with only positive-definite orthogonal spectators cannot have the required one-dimensional full-star relation.

Strict contradiction

Once triangles receive two and quadrilaterals one, every k-gonal two-face has total weight at least $5-k$; nonnegativity suffices for $k \ge 5$. Summing over faces gives total vertex weight at least $5\,f_2 - \binom{n}{2} f_0$.

For a simple compact polytope, the symmetric nonnegative h-vector expresses f_2/f_0 as a weighted average of $\binom{n-i}{2}+\binom{i}{2}$. Its minimum produces the thresholds in the table. The inequality is strict because $h_0=h_n=1$ and the extreme paired coefficient is larger than the minimum. Thus the average vertex weight must exceed 60, 65 or 143/2 in dimensions 25, 26 or 27, respectively. The independently verified upper bounds contradict each case.

This composition is a source-bound finite verification plus the stated all-size geometric reductions, not an enumeration of global polytopes. It makes no priority claim. Complete-mode output exposes the precise ground field, excluded dimensions, upper bound and completion flag so that a separate all-field theorem cannot mistake a partial rule for complete coverage.

```
python3 scripts/fresh_sqrt5_dimension24_coverage_20261015.py
python3 scripts/fresh_sqrt5_dimension24_coverage_20261015.py --verify
```

The final certificate directly checks 2661 zero-touch rows, with dimension-feasible payment counts 25: 18330, 26: 20372, 27: 22565. It verifies 471 literal maps and 70 recursive manifests containing 568 source-hash references. Generation, saved replay, syntax and local-link checks pass.

Source 15

Source path: `results/fresh_sqrt5_dimension27_coverage_20261015.md`

SHA256: db165647f24e8ab1414a29809ee98a6cfc97141f7cacd10339a088c67d7cf5f1

Complete compact Q($\sqrt{5}$) dimension-27 coverage

The binder source and complete certificate compose the frozen, independently reviewed arithmetic queues with direct exact face payments. Generation and saved replay pass. Its `--stage` mode remains a partial diagnostic that writes no data.

The certified conclusion is precise: every **compact arithmetic hyperbolic Coxeter polytope with ground field exactly Q($\sqrt{5}$)** has dimension at most 27. The known compact dimension bound 29 reduces this claim to excluding dimensions 28 and 29. Other ground fields and noncompact chambers are outside this binder. The argument is a composition of reviewed face reductions and exact weights, not an enumeration of global polytopes or a claim of priority.

A fixed rule on full vertex diagrams

Use the frozen cross-weight rule and its separate implementation review. A pair at distance one or two in a spherical vertex component has base weight one. An entire H4 also receives a unit on its 5-edge and on its endpoint pair. The nonnegative connected-pair bonuses depend on the entire component and the number of entire A1 components in the full vertex, capped at two. One additional disconnected-pair bonus, 7/9, joins the unique entire A1 to a specified rooted E7 orbit, only when the full vertex contains exactly one A1.

The unique-A1 condition makes this last contribution additive over E7 components: each root in the specified orbit makes exactly one pair with that A1. There is no uncounted interaction between components. The separate review reconstructs all rooted-node multiplicities, all 92 finite component types through rank 29, the exact primal/dual certificate, and a different exact forest dynamic program. The binder recalculates the component totals and the full forest maxima from the literal rule.

The rule preserves the **weaker H4-only triangle baseline**. It does not require the optional extra unit on an I2(5) or H3 5-edge from the earlier triangle discussion. All 76 minimal Lannér triangle markings are checked directly with the H4-only baseline and receive at least two. Existing paths remain in any spherical extension, and an H4 cannot be a proper connected subdiagram of a larger spherical diagram. These are the extension-stability facts in the independent triangle review. Nonnegative new bonuses cannot reduce those payments.

Exhaustive reduction of an unpaid quadrilateral

A marked distance-one or distance-two corner already pays one, persistently under further white extensions. Hence an unpaid quadrilateral satisfies the no-short-path premise of the complete 5,440-support census and its independent traversal. It is unnecessary to impose every unrestricted 8,298-class union as a new weight constraint: the current rule has never decreased its short-path baseline.

For completeness, the geometric reduction uses the full compact polytope. The common white support W is spherical, and W plus either adjacent black pair is a finite corner. Each opposite-black extension is indefinite. A spherical extension would force the opposite facets to meet; an affine extension is excluded by compactness. Each chosen minimal indefinite piece contains both of its opposite black nodes and belongs to the complete arithmetic Lannér catalogue. The enumeration includes all shared whites and all finite cross contacts. It does not assume that the union is the entire face star.

The 1,180 marked union classes pass through the reviewed principal arithmetic filters: 784 fail the field or signature conditions, leaving 362 finite-label cores and 34 dotted families. The contact review then accounts for all 396 survivors:

Disposition	Classes	Remaining obligation
Persistent entire-H4 payment	84	Direct baseline check
Certified affine subdiagram	2	Compactness exclusion
Singular finite cores with no unpaid contact	112	Entire-component payments
Nonsingular finite cores with no unpaid contact	44	Full-star rank exclusion
Nonsingular finite cores with contacts	130	All 458 first extensions
Dotted families	24	Complete parameter and attachment queues

"No unpaid contact" is conditional sealing. A first touching white root either creates a persistent short-path/H4 payment or occurs in the complete enumerated domain. If all unpaid profiles are excluded, every remaining white root is orthogonal to the core. Its marked components are then entire; arbitrary orthogonal white components remain possible. This is why every payment below is checked with all dimension-feasible spectator A1 counts. A sealed nonsingular core plus positive-definite orthogonal whites stays nonsingular, contrary to the full quadrilateral star's one-dimensional relation.

The finite branches

The complete first-extension containment census partitions the 458 accepted first extensions into 86 singular rows containing an original sealed subcore, 61 further singular rows, and 311 nonsingular rows. The binder reconstructs all 458 literal extensions and their partition. It checks every saved sealed-subcore map and directly evaluates **all 147 singular first-extension rows** with the current rule; it never assumes an old numerical payment transfers to changed weights.

The 61 singular rows have 51 marked representatives. Their independently reviewed complete contact domains give 46 sealed states and five states with seven possible touching continuations. The binder checks all 61 representative maps and all 51 zero-touch payments under the current rule. The 17-state closure and review close the seven touching continuations, together with sixteen auxiliary longer-chain starts. Six states contain known sealed subcores and eleven are newly sealed. All seventeen zero-touch rows, all seven continuation maps, and all saved subcore maps are checked again. Thus the earlier 174-row catalogue is used through explicit current-rule evaluations and structural sealing, not through an assertion that its old weight table is unchanged.

The remaining 311 nonsingular first extensions seed the dimension-specific finite queue. A separately accelerated exact continuation starts from the immutable independently reviewed 1,433-state batch; the original implementation remains unchanged. Its final export and separate complete independent review both pass generation and fresh saved replay. The terminal queue contains 5,172 states and 2,070 singular zero-touch obligations, all checked here. The earlier batch review is explicitly partial provenance, not a claim of exhaustion. Every singular state is a zero-touch obligation, including expanded states and states already paid under older rules. Every nonsingular no-touch state is rank-excluded. A touching state continues through an exact complete one-white domain unless it creates a persistent baseline payment or violates a necessary arithmetic condition.

The boundary of 31 star nodes is geometric for dimensions at most 29. A full n-dimensional quadrilateral star has n+2 roots and rank n+1. A 31-node nonsingular core cannot fit dimension 29; a singular core at that boundary is retained and its zero-touch payment is checked. This is not an experimental cutoff. The queue orders touching nodes by their first contact with the growing core; components orthogonal to it are spectators. Intermediate singular states are not omitted merely because they admit further touching roots.

The dotted branches

The independent parameter review covers all 24 residual families: the exact physical upper bounds, conjugate interval and arithmetic integrality reduce them to 532 tested integer candidates and 32 admissible parameters. No floating-point distance sample is used. Their independently reviewed dimension-29 queue has 438 complete states and 264 singular zero-touch obligations: 250 sealed, four expanded and ten at the dimension boundary.

All 264 literal rows are evaluated directly under the current rule, including their dimension-feasible spectator cases. The dotted parameter is retained in each origin record. Although weights depend only on corner diagrams, parameter-specific arithmetic coverage is inherited from the exact parameter and queue certificates. No sealed-only subset is substituted for the full singular obligation list.

Exact weight contradiction

In a zero-touch row with k star nodes, the displayed corner rank is k−2. At ambient dimension n, the external white rank is n−k+2. The binder checks every possible capped A1 count obtained by assigning some of that rank to A1 components and leaving either zero or at least two dimensions for non-A1 components. Thus the boundary cases do not assume that all three contexts are always realizable.

The exact maximum total vertex weights are:

Dimension	No A1	Exactly one A1	At least two A1	Overall
28	230/3	230/3	230/3	230/3
29	239/3	84	236/3	84

With all quadrilateral cases paid, every k-gonal two-face receives at least 5−k: triangles receive two, quadrilaterals one, and larger faces need only nonnegative weights. Double-counting gives total weight at least $5\,f_2 - \binom{n}{2} f_0$.

For a simple compact polytope, the symmetric nonnegative h-vector writes f_2/f_0 as an average of the paired coefficients $\binom{n-i}{2}+\binom{i}{2})/2$. Their minima are 91 at n=28 and 98 at n=29. The ratio is strictly greater than these minima because $h_0=h_n=1$ and the extreme paired coefficient is larger. Hence the average vertex weight is strictly greater than 77 at dimension 28 and strictly greater than 84 at dimension 29. Both contradict the displayed upper bounds. This is the same published weighted-face/h-vector framework used in the independent cubic-field proof review.

The complete certificate checks 471 literal maps and 2,661 zero-touch rows in 13,901 dimension-specific payment cases. It combines complete branch coverage, direct current-rule payments and this strict inequality. Generation and saved replay pass, with 60 recursive manifests and 477 source-hash references verified.

Source 16

Source path: `results/fresh_sqrt5_dotted_parameters_20261015.md`

SHA256: 75d5125b6a104f3f9a208d7bdd74f8bcb244b9a433bca21c572a15c98bc386ff

Complete arithmetic distance domains for the remaining dotted cores

The exact checker reduces all 24 unpaid minimal dotted families to **32 discrete parameter cases**: eighteen singular cores and fourteen nonsingular cores. Its certificate exhausts 532 field integers in explicit bounded embedding boxes. This parameter reduction allows arbitrary subsequent attachments; it is not a distance cutoff.

Let the scaled Gram diagonals of the dotted endpoints be d0,d2, and their inner product -x. In every seed, the common principal matrix B obtained by deleting these endpoints is nonsingular with one negative direction. The two-by-two Schur residual has diagonals a,b and off-diagonal -x-c. Any physical extension with only one negative direction requires this residual to be positive semidefinite. Consequently (x+c)^2≤ab, and

$t = 4x^2/(d0d2) \le 8(ab+c^2)/(d0d2).$

The displayed inequality follows from $x^2 \le 2(x+c)^2+2c^2$. Exact rounding of the right side gives upper bounds between 9 and 76. Since the walls are disjoint, t>4. Arithmeticity requires the two-edge cyclic product t to be an algebraic integer in Q(sqrt(5)). Conjugate positive semidefiniteness requires $0 \le t' \le 4$.

Write $t=A+B\varphi$, with $\varphi=(1+\sqrt{5})/2$ and A,B integers. The embedding bounds imply B>0, A≥0, and both coefficients are at most the integer upper bound. The checker enumerates that finite rectangle and tests both embedding inequalities exactly. The endpoints t'=0,4 cannot actually occur with t>4, because conjugation is injective and fixes rational numbers.

Of the 532 candidates, 365 fail the dotted cyclic-field condition, 107 force two physical negative directions, and 28 fail conjugate positive semidefiniteness. All remaining 32 pass every simple-cycle integrality test. They are necessary principal cores, not asserted realized faces. The two minimal classes 6 and 7 have no surviving arithmetic distance.

Generation and saved replay pass. Independent review of this new reduction is pending. Attachment closure and compatible face weights are still needed before it can contribute to an improved dimension theorem.

```
python3 scripts/fresh_sqrt5_dotted_parameters_20261015.py --verify
```

Source 17

Source path: `results/fresh_sqrt5_dotted_rank29_exhaustion_20261015.md`

SHA256: e76044cb78a35c0cbcf2baf658590b6925403132e9034e305caa33e143594a86

Complete dotted attachment catalogue through dimension29

The checker exhausts the attachment queue of all 32 necessary arithmetic parameter seeds from the complete dotted-distance reduction. Its certificate has **438 marked states and no pending state**. It reduces the dotted part of the dimension-28/29 argument to **264 explicit face-weight obligations**. Those weights have not yet been verified; this is not an improved dimension theorem.

Processed outcome	States
Singular state sealed against further unpaid contact	250
Singular state with a retained next contact	4
Nonsingular state with retained next contacts	156
Sealed nonsingular state excluded by full-star rank	13
Nonsingular31-root state excluded by ambient rank	5
Singular31-root state at the dimension boundary	10

The inherited arithmetic upper bound is29. A full quadrilateral star in dimension d has d+2 roots and Gram rank d+1. Therefore a relevant star has at most31 roots. At31 roots, a nonsingular Gram matrix already exceeds the allowed ambient rank; a singular one is a possible zero-contact weight case in dimension29. This is an explicit restriction to dimensions at most29, not an arbitrary-rank sealing assertion.

Below that boundary, every possible nonzero next-white contact is generated in the full finite-angle alphabet, subject only to the reviewed four finite corners and no-short-path condition. Exact field, physical/conjugate inertia and cyclic-integrality tests leave422 accepted child occurrences. Every one has a saved literal marked isomorphism to a queue state. The remaining trials give3,457 two-negative exclusions,854 finite-cycle field exclusions,334 dotted cycle field exclusions,66 conjugate exclusions, and13 persistent H4 payments. The checker also has a nonnegative-root-dependence rejection, but no trial needed that branch. It makes no negative assertion based on a partial search for affine subdiagrams.

Deduplication uses exact full-label marked graph isomorphisms within invariant buckets, retaining the squared dotted distance as part of the identity. It does not identify geometrically different dotted parameters. There is no sampling, and the empty queue is checked directly.

In an unpaid actual full star, any nonorthogonal external white component can be added from its first contact with the core, then along its connected white edges. Thus it follows a retained branch. A sealed nonsingular core cannot be completed by orthogonal spectators: their positive-definite Gram block would leave the whole star nonsingular. A singular state can instead stop with only orthogonal spectators. Each of the250 sealed singular states, four nonsealed singular starts, and ten boundary states must therefore receive the required face payment, with spectator rank limited by the same dimension bound. Proving those264 obligations pays every previously unresolved dotted family in the targeted dimensions.

Generation and saved replay pass. Independent review is being performed separately. All finite-label attachment families and the compatible global vertex budget remain separate obligations.

```
python3 scripts/fresh_sqrt5_dotted_rank29_exhaustion_20261015.py --verify
```

Source 18

Source path: `results/fresh_sqrt5_hub_count_coverage_20261016.md`

SHA256: c56e5447cb59fe7c422677231e871345eb1d6b0f69b971d0ffbc97e53749f2a2

Complete Q($\sqrt{5}$) coverage with capped hub counts

The new internally reviewed computer-assisted proof candidate excludes dimension 23 for compact arithmetic Coxeter polytopes with ground field Q($\sqrt{5}$). Together with the frozen exclusions of dimensions 24–29 and the published compact ceiling 29, this gives **dimension at most 22** for that field. Generation and saved replay of the binder and its bound-22 certificate pass. External mathematical validation and priority remain open.

The new rule and separate review preserve all fixed short-path and H4 payments. Its additional bonuses use the entire A1 count and the I2(5) count capped at two. Its exact vertex maximum is $485211/10000 = 48.5211$. The denominator in a cross angle remains the actual source count, irrespective of the capped coefficient context.

Direct coverage under the changed rule

The binder imports the frozen complete arithmetic classifications and literal reduction maps from the previous coverage. It reevaluates every older sealed branch, singular first extension, singular continuation, dotted stopping state, and complete finite-queue singular state with the new rule. No previous numerical payment is assumed to transfer. All 2,661 zero-touch rows remain in the output; 1,764 fit dimension 23 and receive 52,029 common spectator checks. The other 897 are explicitly excluded at this target dimension by corner rank. The binder verifies 471 literal maps and 76 recursive manifests with 872 hash references.

A k-node core has external rank 25−k. Every common external pair `(e,f)` of A1 and I2(5) counts is tested when `e+2f≤25−k` and the leftover rank is zero or at least two. A2 and A3 components suffice to realize every leftover rank at least two without changing either count. Each corner adds its own core counts before applying the cap or either denominator. This is a conservative enumeration; no arithmetic realization of each forest is assumed.

Structural completeness is inherited from the independently reviewed finite and dotted queues, their complete one-white domains and the earlier conditional-sealing proofs. A first touching white root either creates a persistent baseline payment or follows a retained branch. Orthogonal white components are precisely the spectators above. All singular stopping alternatives are checked, including expanded states. Nonsingular no-touch states are rank-excluded. The justified 31-node limit covers every target dimension at most 29, including 23.

The strict contradiction

The 76 triangle markings still receive at least two. Every quadrilateral receives at least one, and all larger-face weights are nonnegative. Thus the average vertex weight is at least $5\,f_2/f_0 - \binom{23}{2}$. The symmetric nonnegative h-vector of a simple compact polytope gives $f_2/f_0 > 121/2$; strictness follows from $h_0 = h_{23} = 1$ and the larger extreme paired coefficient. The average vertex weight is consequently strictly greater than $5 \cdot 121/2 - 253 = 99/2$. This contradicts the exact upper bound 48.5211.

The certificate explicitly combines this new exclusion with the complete previous exclusions of dimensions 24–29. Other arithmetic fields and the noncompact case are handled in the all-field composition.

```
python3 scripts/fresh_sqrt5_hub_count_coverage_20261016.py --dimension 23 --hub-cap 2
python3 scripts/fresh_sqrt5_hub_count_coverage_20261016.py --dimension 23 --hub-cap 2 --verify
```

Source 19

Source path: `results/fresh_sqrt5_lanner_catalogue_20261015.md`

SHA256: 47a5aca7018ba5ec918a78079db376685d2f0246fc7ab876511f0143a9cbd046

Complete arithmetic Lannér input over Q(sqrt5)

The exact checker and certificate give the complete local Lannér catalogue compatible with compact arithmetic ground field $K = \mathbb{Q}(\sqrt{5})$: an all-parameter dotted family, **12 triangles, six order-four types, and four order-five types**. No ambient quadrilateral classification or dimension improvement is claimed.

Set $\phi = (1+\sqrt{5})/2$, so $O_K = \mathbb{Z}[\phi]$ and $\phi' = 1-\phi$. All roots are normalized to square 2. A finite bond m has Gram entry $-2\cos(\pi/m)$. The arithmetic criterion imposes integral cyclic products in K, a totally real entry field, and positive semidefinite conjugate Grams at embeddings nontrivial on K. For nonsingular principal blocks this requires positive definiteness. These are the published conditions recalled in Nikulin, section 2, with the opposite overall sign convention there. The maximum Lannér order five and the interpretation as minimal indefinite diagrams are the published inputs from Vinberg, section 2.4.

All finite labels and dotted pairs

The two-cycle is $4\cos^2(\pi/m) = 2 + 2\cos(2\pi/m) \in K$. Its real cyclotomic degree is at most two, giving candidates $2, 3, 4, 5, 6, 8, 10, 12$. Labels 8 and 12 generate Q(sqrt2) and Q(sqrt3), respectively, so the complete finite alphabet is

$$\boxed{2, 3, 4, 5, 6, 10}.$$

There is no finite-label cutoff assumption.

For an order-two Lannér block write its off-diagonal entry as $-\sqrt{t}$. The complete arithmetic conditions are

$$\boxed{t \in \mathbb{Z}[\phi], \quad t > 4, \quad 0 < t' < 4.}$$

These imply total reality of the radical field and conjugate positive definiteness, and are also necessary. Endpoint equality cannot occur because an embedding cannot send the nonzero algebraic number t or t minus 4 to zero. The example $t = 3+\phi$ is verified. Dotted distances are not discretized or bounded by a search cutoff.

Exactly 12 triangle types

The relevant contact squares are

$$0, 1, 2, \phi+1, 3, \phi+2$$

for labels $2, 3, 4, 5, 6, 10$. The classes of 2, 3, and $\phi+2$ are independent in K^*/K^{*2}. Indeed 2, 3, and 6 are not squares in K: writing a square as $(a+b\sqrt{5})^2$ forces $ab = 0$, and neither d nor d/5 is a rational square for $d = 2, 3, 6$. The norm of $\phi+2$ is 5, so multiplying it by any product of 2 and 3 still has nonsquare rational norm. Thus a nonzero triangle cycle belongs to K exactly when the counts of labels 4, 6, and 10 are each even. For a right triangle the cycle vanishes and this parity condition is unnecessary.

The checker handles all 56 unordered label triples. Physical hyperbolicity is $1/p + 1/q + 1/r < 1$. If the three contact magnitudes are a,b,c, the determinant is $8 - 2(a^2+b^2+c^2) - 2abc$. Each conjugate proper two-node block is positive; the remaining test is the exact positivity of the conjugate determinant.

Labels	Norm-two determinant
2,3,10	$2-2\phi$
2,4,5	$2-2\phi$
2,4,10	-2ϕ
2,5,5	$4-4\phi$
2,5,6	-2ϕ
2,5,10	$2-4\phi$
2,10,10	-4ϕ
3,3,5	$2-4\phi$
3,5,5	-6ϕ
4,4,5	$-2-6\phi$
5,5,5	-10ϕ
5,10,10	$-4-12\phi$

All listed conjugate determinants are positive. The certificate records every rejected triple and its exact rejection reason. All cyclic products that pass field membership are integral, since the individual finite-contact entries are algebraic integers. The required radical fields are totally real: 2 and 3 are positive, and both conjugates of $\phi+2$ are positive. Different choices of radical signs over the same ground-field embedding are vertex sign switches, because their cycle products agree; testing one resulting Gram suffices.

Orders four and five

A connected diagram of order at least four with a bond at least 6 contains a proper three-node path with that bond and another nonright bond. Its relaxed unit-normal determinant is at most $1-\cos^2(\pi/6)-\cos^2(\pi/3)=0$; an extra bond only decreases it. Parallel and dotted bonds already violate proper two-node positivity. Consequently every bond here is 2, 3, 4, or 5.

A Lannér graph is connected. If it contains a cycle, the cycle must span all nodes: an all-3 cycle already has a nonpositive all-ones quadratic form, and larger labels or chords decrease that value. A proper cycle would contradict proper-subdiagram positivity. A chord creates a shorter proper cycle. Thus the only graph topologies to consider are **trees and spanning chordless cycles**.

The checker enumerates all labelled trees using Prüfer words and all labelled cycles, with every assignment of labels 3,4,5 to their edges. This gives 675 order-four and 13,041 order-five candidates. The finite proper-diagram filter is independently checked against exact Sylvester minors for all 64 and 4,096 labelled rank-three and rank-four graphs. Full determinants are computed in the exact basis $(1,\phi,\sqrt{2},\phi\sqrt{2})$. A tree has no nontrivial cycle; a spanning cycle has its product in K precisely when its number of 4-bonds is even. Every accepted diagram is also checked against all proper physical and conjugate principal minors.

Order	Diagram	Norm-two determinant
4	Three-arm star with labels 3,3,5	$4-4\phi$
4	Path 3–5–3	$5-4\phi$
4	Path 4–3–5	$2-2\phi$
4	Path 5–3–5	$6-5\phi$
4	Four-cycle with one 5, otherwise 3	$2-5\phi$
4	Four-cycle with opposite 5s, otherwise 3	$1-7\phi$
5	D5 tree with a 5 at the terminal edge of its long arm	$4-4\phi$
5	Path 5–3–3–3	$6-4\phi$
5	Path 4–3–3–5	$2-2\phi$
5	Path 5–3–3–5	$8-6\phi$

There are 78 and 360 labelled arithmetic survivors, respectively. Before the arithmetic tests the search recovers nine geometric order-four and five geometric order-five classes. The older repository census restricted to entries $0, -1, -\phi$ is a proper subfamily: it does not cover the new triangle labels 4,6,10 or the two displayed paths containing a 4-bond. This package closes that local-input gap, without claiming novelty for the underlying classical lists.

Scope and reproduction

Every Lannér subdiagram in a compact arithmetic Q(sqrt5) chamber must belong to this catalogue. Each listed finite type satisfies the local arithmetic tests. The list does not classify unions of opposite Lannér pieces, shared white vertices, cross contacts, or arbitrarily large external white components in a quadrilateral star. Unlike the Q(sqrt2) order-four/five list, this field permits several tree types, so a proof using only cycle pieces cannot be transferred. The existing E7/E8 marked-triangle restrictions likewise do not replace the missing global quadrilateral analysis.

```
python3 scripts/fresh_sqrt5_lanner_catalogue_20261015.py
python3 scripts/fresh_sqrt5_lanner_catalogue_20261015.py --verify
```

Primary generation and saved replay pass, along with source hashes, syntax, and local links. Independent review has been assigned separately. No dimension bound is asserted here.

Source 20

Source path: `results/fresh_sqrt5_lanner_catalogue_20261015_review.md`

SHA256: dbdbd818f43a3a8857df4ee12298a2865337a911a3e02a8090a3f397622e455a

Independent review of the Q(sqrt5) Lannér catalogue

PASS. The independent checker and saved certificate reproduce the complete finite catalogue in the primary report: 12 triangles, six order-four classes and four order-five classes. The dotted order-two family remains symbolic, with squared norm-two contact $t \in \mathbb{Z}[\phi]$, $t > 4$, and $0 < t' < 4$. This review does not assert a new dimension bound.

The implementation imports no producer or other project code. It uses integer arithmetic in the basis $1, \sqrt{2}, \sqrt{5}, \sqrt{10}$, doubled Gram matrices, permutation determinants, exact nested-quadratic signs and Sylvester minors. In particular, it does not reuse the producer's finite Dynkin classifier. It independently tests all 64 rank-three and 4,096 rank-four label matrices, obtaining 25 and 182 positive patterns. It enumerates higher-order graph supports by connected edge masks, independently of the producer's Prüfer-word construction.

All 56 unordered finite-label triples and their rejection reasons agree. All 675 order-four and 13,041 order-five tree/cycle assignments agree, including every geometric class, determinant, conjugate determinant, field rejection and positive-definiteness rejection. The labelled arithmetic survivor counts are 78 and 360. Determinants are converted independently to the producer's golden-ratio basis before comparison.

The all-label and topology reductions were separately source/proof reviewed. The two-cycle field condition gives exactly labels $2, 3, 4, 5, 6, 10$. The square classes of $2, 3, \phi + 2$ are independent over $\mathbb{Q}(\sqrt{5})$, so the stated cycle parity test is complete. A connected Lannér diagram of order four or five cannot contain a bond at least six, by a proper three-node positive-definiteness test. Any cycle must be spanning and chordless; otherwise a proper cycle already has a nonpositive all-ones quadratic form. Thus the enumerated trees and spanning cycles exhaust the relevant supports. Conjugate positivity, cyclic field membership and integrality are checked with the stated normalization. The maximum Lannér order and arithmetic Gram criterion remain the published theorem dependencies identified in the primary report; this review does not reprove those external theorems.

Generation and saved replay pass. The certificate binds the final primary source and JSON, plus the independent source; the primary prose is not hash-bound. Full ambient attachments, opposite-piece unions, quadrilateral stars and any Q(sqrt5) dimension improvement remain outside this review.

```
python3 scripts/fresh_sqrt5_lanner_catalogue_20261015_review.py --verify
```

Source 21

Source path: `results/fresh_sqrt5_nonsingular_rank29_fast_20261015_review.md`

SHA256: 5746f6dad2cee688ad58b8d19a54ab0ae0187b8dc755ebcde135e2d11bf00f9b

Independent terminal finite-attachment review

PASS: independent generation and fresh saved replay. The independent checker and its certificate reconstruct the complete frozen queue: 5,172 states, 51,599 contact profiles, 10,085 literal seed/child maps, and 8,280 corner patterns. All 2,070 singular states remain explicit weight obligations. The source and certificate are final and frozen. This review does not itself prove a dimension bound.

The arithmetic implementation imports neither the fast producer nor its tuple-field, minimal-gauge or white-leaf elimination routines. It uses the older separately reviewed Fraction implementation in the basis `1,sqrt(5)`, its tree-product Gram gauge, and the original forward finite Dynkin domain enumeration. This contrasts with the fast producer's reverse finite-leaf domains and minimal-gauge sparse elimination. Previously reviewed graph typing, marked-map checking and H4-payment helpers are shared and identified as such; their old classifications are not re-proved here.

Independent matrix identities

For an `N`-node core choose the finite corner consisting of all white roots and black roots0,1. In block notation write

`M = [[A,V],[V^t,B]]`, with `A` of order `N-2`.

The checker performs direct Gauss–Jordan elimination on `A`. Every pivot is strictly positive at both embeddings, and the final inverse is checked by literal multiplication. Thus both full inertias are those of `S = B-V^t A^-1 V`, plus `N-2` positive directions. Only this2×2 matrix uses the inherited general inertia routine. The usual block inverse gives a cached inverse of `M` when nonsingular. If `S` has corank one, its kernel lifts by `(-A^-1 V y,y)`; it is checked against every row of `M`. Deleting one suitable outside node gives a nonsingular Lorentz ground and a cached grounded inverse. No white-leaf elimination is used.

The tree-product gauge may change when a new root is attached. The old block `B0` of the enlarged Gram is therefore checked coefficientwise to be

`B0 = kappa R M R`,

where `R` is diagonal with physically positive entries and `kappa` is positive at both embeddings. Connectivity of every retained core is checked explicitly; the diagonal factors are propagated along its actual edges and then verified on every matrix entry. After this congruence and a positive scalar division, the new block has old part `M`, contact vector `u = (kappa R)^-1 v`, and new square `d/kappa`.

For a singular old core, its kernel must annihilate `u`. The reviewer checks this directly, independently of the producer's encoded relation products. Otherwise the profile is excluded. If the relation is respected, the new scalar Schur complement is

`s = d/kappa - u_ground^t M_ground^-1 u_ground`.

For a nonsingular core the same formula uses the full inverse. Its signs at both embeddings give the new inertias exactly. When a new kernel is created, `(-M^-1 u,1)` is transported back through the diagonal congruence and checked against the literal enlarged Gram. Its physical signs test the nonnegative-root-dependence obstruction. A transported inherited kernel has the same physical signs because all diagonal factors are positive.

The checker matched nine original-engine states ranging through20 nodes, including a28-profile case, and separately reconstructed a31-node member of the frozen subdivision families with inertia `(29,1,1)` and conjugate inertia `(30,0,1)`. These are development checks, not a substitute for the terminal replay.

Generation can start from the immutable completed3,473-core snapshot while the producer continues. Those arithmetic results are retained only in the same process, keyed by the complete literal record hash. Final use requires exact equality with the terminal record; no external cache or assumed producer decision is trusted. The saved replay disallows this prefetch option and reconstructs every arithmetic case afresh.

Coverage and database contract

The terminal run is required to have an empty queue, every result present, and a cursor equal to its total state count. The independent checker reconstructs all311 literal seed maps, every complete forward contact domain, every rejection or accepted child, all literal child maps and reachability. It directly compares the preserved original-engine 2,007-core snapshot with the corresponding final rows and decisions; gauge-specific relation coordinates and traversal counters are not mistaken for invariant data.

Every singular state, including one with accepted children or at the 31-node boundary, must appear as a zero-further-contact obligation. All four full corner forests, marked component orbits, existingA1 counts and dimension-specific spectator rank budgets are recomputed. No older weight payment is inherited as a reason to drop a case. The final database is opened read-only and checked for integrity, literal rows/results, cursor, seed records and its bound hash. All counters and obligation indices are reconstructed from the records.

The terminal status counts are 2,760 expanded nonsingular states, 132 sealed nonsingular rank exclusions, 210 nonsingular states excluded at the dimension boundary, 1,871 sealed singular obligations, 43 expanded singular obligations, and 156 singular boundary obligations. The queue has no pending state. There are 9,774 accepted child profiles; the other 41,825 profiles are excluded by exact field, inertia or inherited-relation tests, or paid by the already reviewed entire-H4 rule. The 311 seed maps plus 9,774 child maps account for the full map total.

The final primary source, JSON and database are immutable. The database is cryptographically bound through the primary JSON's explicit SHA256 and checked directly by this reviewer. The primary Markdown report is not bound, so it can receive this review link without changing any mathematical input. The producer's terminal structural audit is distinct from this independent arithmetic replay; a full saved replay of the fast producer is not claimed here.

The dimension boundary is a logical use of the inherited compact arithmetic upper bound29. A quadrilateral star in dimension `d` has `d+2` nodes and Gram rank `d+1`. Thus a nonsingular31-node core is impossible for `d≤29`, while a singular31-node core remains a weight obligation. If there is no further touching white root, all outside white components are positive definite and orthogonal to the displayed core; they cannot rescue a nonsingular full-star rank, and they must be included in a singular state's possible forest-sensitive payment. Complete one-white continuation is valid because any touching outside component has a first root adjacent to the current core.

The stored spectator budgets are for dimensions 28 and 29. The same queue is a necessary superset for a smaller ambient dimension: retain only states with `N≤d+2` and recompute the spectator budget `d+2−N`. Such a use still requires its own weight-payment checks; the recorded dimension-28/29 budgets are not substitutes for that calculation.

This review does not establish geometric realization of the necessary principal stars, pay their weights, cover the other finite/dotted seed families, or independently prove a global dimension bound. Those are separate inputs to the final coverage certificate. Independent generation and the fresh saved replay both pass on the frozen terminal data. The saved replay rebuilt all arithmetic without the generation process's prefetch cache. Source syntax, recursive manifests and local report links also pass.

Source 22

Source path: `results/fresh_sqrt5_quad_exhaustion_20261015.md`

SHA256: 8f5b8b3088f430c546aab790cee890782f32cbca1f9503a1bfeaa6d0b5b2a5b5

Complete minimal-pair unions over Q(sqrt5)

The enumerator and certificate exhaust the union of two opposite minimal Lannér pieces in a compact arithmetic Q(sqrt5) quadrilateral star having **no marked corner pair at graph distance at most two**. There are **5,440 distinct labelled union supports and 1,180 marked isomorphism classes**. This is a complete necessary local list; it does not assert that any listed support extends to a full face star or a compact polytope.

The finite alphabet and minimal pieces come from the complete catalogue and its independent review. They include all labels 2,3,4,5,6,10, the symbolic dotted order-two family, all 12 triangle types, six order-four types and four order-five types. Dotted contacts are represented by label zero and remain undetermined.

Let the cyclic black labels be 0,1,2,3 and let W be the white support of a quadrilateral. Its four corners have spherical diagrams W plus one black node from each opposite pair. The diagrams W+{0,2} and W+{1,3} are indefinite; choose minimal indefinite pieces in them. Each chosen piece contains both of its opposite black nodes, because deleting either leaves a subdiagram of a spherical corner. Compactness excludes affine alternatives, so the chosen pieces belong to the Lannér catalogue.

For each piece, black names are fixed while white permutations are quotiented. All placements of black nodes in every catalogue graph are retained. This gives 153 representatives: one dotted pair, 49 marked triangles, 40 order-four pieces and 63 order-five pieces. The enumerator tries every injective shared-white identification, including disjoint white sets. Swapping the two opposite pairs justifies using unordered piece indices; no polytope automorphism is assumed.

Every pair not already fixed by a piece is assigned every finite label in the complete alphabet. These free pairs belong to at least one actual corner, so they cannot be dotted or parallel. A direct adjacent-black contact, or a common white neighbor of adjacent black nodes, violates the no-distance-two premise. All four induced corner Grams must be positive definite. The primary checker uses exact forest Schur elimination over Q(phi), including isolated I2(6) and I2(10) factors. Cycles have a nonpositive quadratic witness, while a bond at least six cannot meet another nonright bond in a spherical component. Assigning remaining contacts right is the weakest acute Gram, making early failures safe monotone prunes. If the two chosen indefinite pieces are disjoint and orthogonal, their union has at least two negative directions and is excluded.

The exhaustive traversal has no cutoff. It checks 127,875 shared-white identifications, of which 12,452 pass the initial corner and short-path tests, and visits 207,319 DFS nodes. It rejects 11,781 orthogonal-piece completions and retains 5,502 completions before literal duplicate removal. The 5,440 distinct supports have six to ten nodes. Every isomorphism class includes explicit full label-preserving maps that preserve black/white colors and the two opposite pairs.

The separate coverage replay uses finite Dynkin recognition instead of the primary Schur arithmetic, a different shared-white generator, reversed contact order and an independent canonicalization. Its complete 196,742-node traversal recovers exactly the same 5,440 supports and 1,180 classes. Generation and saved replay pass for both implementations.

Outside white vertices are not enumerated here. Cyclic field membership of the union, conjugate positivity, Lorentz inertia, full-star rank, arbitrary external white components, and the extra triangle-weight corrections require separate arguments. In particular, this list alone gives no new Q(sqrt5) dimension bound. The separately produced arithmetic filters may remove supports, and some surviving local corners already earn an entire-H4 correction; neither fact is silently used to restrict this enumeration.

```
python3 scripts/fresh_sqrt5_quad_exhaustion_20261015.py --verify
python3 scripts/fresh_sqrt5_quad_exhaustion_20261015_incidence_review.py --verify
```

The optional running snapshot stores only completed outer-piece ranges. `--resume` checks its source hashes and continues at the next outer piece; an interrupted current piece is recomputed without losing later siblings. The final certificate above

is the authoritative completed result.